

COMP9444 Assignment 1

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Mon 7 July 2025

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Part 1: Japanese Character Recognition

Question 1

Final Accuracy & Confusion Matrix:

```
[[763.  6.  8. 12. 30. 65.  2. 62. 33. 19.]
 [  7. 665. 107. 19. 32. 23. 57. 14. 25. 51.]
 [  7. 62. 693. 25. 25. 20. 47. 37. 46. 38.]
 [  5. 34. 60. 759. 15. 56. 13. 18. 28. 12.]
 [ 58. 51. 78. 20. 627. 20. 33. 35. 21. 57.]
 [  7. 28. 125. 17. 19. 725. 28.  8. 34.  9.]
 [  5. 24. 149. 10. 24. 22. 723. 20.  9. 14.]
 [ 16. 28. 27. 11. 86. 19. 50. 625. 89. 49.]
 [ 11. 37. 97. 39.  7. 29. 44.  6. 709. 21.]
 [  9. 51. 91.  4. 52. 30. 18. 31. 41. 673.]]
```

Test set: Average loss: 1.0104, Accuracy: 6962/10000 (70%)

Question 2

Final Accuracy & Confusion Matrix:

```
[[856.  3.  2.  5. 28. 23.  3. 38. 36.  6.]
 [  5. 818. 31.  4. 18. 10. 61.  2. 20. 31.]
 [  7. 10. 842. 45. 11. 16. 25. 12. 19. 13.]
 [  2. 10. 30. 920.  2. 12.  5.  4.  6.  9.]
 [ 42. 26. 21.  7. 806. 11. 33. 16. 20. 18.]
 [ 11. 14. 85. 13.  9. 816. 24.  3. 17.  8.]
 [  3. 15. 50. 10. 15.  3. 889.  9.  2.  4.]
 [ 14. 15. 18.  2. 21. 15. 25. 839. 24. 27.]
 [  8. 24. 28. 50.  4.  9. 31.  3. 835.  8.]
 [  5. 20. 44.  7. 29.  6. 13. 15. 14. 847.]]
```

Test set: Average loss: 0.5061, Accuracy: 8468/10000 (85%)

Calculation of Number of Independent Parameters in Network

```
30 ~ class NetFull(nn.Module):
31     # two fully connected tanh layers followed by log softmax
32 ~     def __init__(self):
33         super(NetFull, self).__init__()
34         n_nodes = 150
35         self.in_to_h1 = torch.nn.Linear(28*28, n_nodes)
36         self.h1_to_out = torch.nn.Linear(n_nodes, 10)
37
38 ~     def forward(self, x):
39         x_flattened = torch.flatten(input=x, start_dim=1) # dim=1
40         net_h1 = self.in_to_h1(x_flattened)
41
42         out_h1 = F.tanh(net_h1)
43         net_out = self.h1_to_out(out_h1)
44
45         output = F.log_softmax(net_out, dim=1)
46         return output
```

NOTE: The model architecture is specified in the image above.

The input is a 28×28 image. Once flattened, this results in $28 \times 28 = 784$ input nodes. The first layer contains 150 hidden nodes, each of which has an associated bias term. Therefore, the connections between the inputs and hidden layer nodes contains a total of $150 \times (784 + 1) = 117,600 + 150 = 117,750$ parameters. Note, the blue term represents the biases.

The output layer contains 10 nodes, each of which has an associated bias term. Recall, the previous (hidden) layer contained 150 nodes, therefore the connections between the hidden and output layers contains a total of $10 \times (150 + 1) = 1,500 + 10 = 1,510$. Note, the blue term represents the biases.

Thus, the total number of independent parameters in this network is:

$$117,750 + 1,510 = 119,260$$

Question 3

Final Accuracy & Confusion Matrix:

```
[ [956.  2.  1.  0. 31.  1.  0.  6.  1.  2.]
[  2. 919.  6.  0. 10.  2. 38.  5.  7. 11.]
[ 11.  3. 881. 53.  8.  9. 22.  4.  4.  5.]
[  0.  0.  10. 964.  2.  6. 10.  2.  3.  3.]
[ 22.  5.  1.  4. 937.  2. 10.  6.  6.  7.]
[  3.  9. 25.  8.  4. 923. 15.  2.  9.  2.]
[  4.  4.  7.  0.  3.  0. 978.  0.  1.  3.]
[  8.  3.  1.  0. 14.  2.  9. 942.  6. 15.]
[  4. 13.  6.  8. 15.  5.  9.  3. 929.  8.]
[  6.  5.  5.  2.  4.  0.  2.  0.  4. 972.]]
```

Test set: Average loss: 0.2185, Accuracy: 9401/10000 (94%)

Calculation of Number of Independent Parameters in Network

```
49 class NetConv(nn.Module):
50     # two convolutional layers and one fully connected layer,
51     # all using relu, followed by log_softmax
52     def __init__(self):
53         super(NetConv, self).__init__()
54         # inputDim = 28
55         # outputDim = (28 - 3)/1 + 1 = 26
56         self.in_to_c1 = torch.nn.Conv2d(in_channels=1, out_channels=20, kernel_size=3, stride=1)
57
58         # inputDim = 26
59         # outputDim = (26 - 4) / 2 + 1 = 12
60         self.pool_c1 = nn.MaxPool2d(kernel_size=4, stride=2)
61
62         # inputDim = 12
63         # outputDim = (12 - 3) / 1 + 1 = 10
64         self.c1_to_c2 = torch.nn.Conv2d(in_channels=20, out_channels=40, kernel_size=3, stride=1)
65
66         # inputDim = 10
67         # outputDim = (10 - 4) / 2 + 1 = 4
68         self.pool_c2 = nn.MaxPool2d(kernel_size=4, stride=2)
69
70         # inputDim = 40*4*4 (flattened)
71         # outputDim set to 144
72         self.c2_to_fc1 = torch.nn.Linear(in_features=40*4*4, out_features=144)
73
74         # inputDim = 144
75         # outputDim set to 10
76         self.fc1_to_out = torch.nn.Linear(in_features=144, out_features=10)
```

NOTE: The model architecture is specified in the image above.

Consider each 2D layer in order:

- Input: $28 \times 28 \times 1$ (grayscale image)
- Convolution Layer 1: $\text{DimOut} = \frac{\text{DimIn} - \text{FilterSize}}{\text{StrideSize}} + 1 = \frac{28-3}{1} + 1 \Rightarrow 26 \times 26$. Pooling Layer 1: $\text{DimOut} = \frac{\text{DimIn} - \text{FilterSize}}{\text{StrideSize}} = \frac{26-4}{2} + 1 \Rightarrow 12 \times 12$
- Convolution layer 2: $\text{DimOut} = \frac{\text{DimIn} - \text{FilterSize}}{\text{StrideSize}} = \frac{12-3}{1} + 1 \Rightarrow 10 \times 10$
- Pooling Layer 2: $\text{DimOut} = \frac{\text{DimIn} - \text{FilterSize}}{\text{StrideSize}} = \frac{10-4}{2} + 1 \Rightarrow 4 \times 4$

There are 20 filters in the first convolution layer. Each of these filters have dimensions $3 \times 3 \times 1$ (grayscale), and thus there are $3 \times 3 \times 1 = 9$ weights per filter. By default, PyTorch convolution layers also contain 1 bias term per filter, resulting in $20 \times (9 + 1) = 180 + 20 = 200$ independent parameters in this layer. Note, the blue term represents the biases.

The pooling layer has no learnable parameters. However, it is important to note that the output is 20 channels, each with a dimension of 12×12 .

There are 40 filters per input channel in the second convolution layer. Each of these filters have a $3 \times 3 \times 20$ dimension, and thus there are $3 \times 3 \times 20 = 180$ parameters per filter. As there are 40 filters containing a bias term each, this results in a total of $40 \times (180 + 1) = 7200 + 40 = 7240$ independent parameters in this layer.

Once again, the second pooling layer has no learnable parameters. However, it is important to note that the output dimensions are 4×4 .

Thus, there are now 40 channels, each with a dimension of 4×4 . After flattening, this results in $40 \times 4 \times 4 = 640$ input nodes for the fully connected (FC) layer. As there are 144 nodes in the FC layer, each with a bias term, this layer has a total of $144 \times (640 + 1) = 92160 + 144 = 92304$ independent parameters in this layer.

Finally, there are 10 nodes in the output layer, each with a bias term. Recall that there were 144 nodes in the previous FC layer, and thus there are $10 \times (144 + 1) = 1440 + 10 = 1450$ independent parameters in this layer.

Thus, the total number of independent parameters in this network is:

$$200 + 7240 + 92304 + 1450 = 101,194$$

Question 4

Model	Accuracy	# Independent Parameters
NetLin	70%	7,850
NetFull	85%	119,260
NetConv	94%	101,194

*Note, the calculation for the number of parameters in NetLin is: $(28 \times 28 \times 1) \times (10 + 1) = 7840 + 10 = 7850$

Relative Accuracy of Each Model

The NetLin model achieves an accuracy of 70%. This is significantly lower than the accuracy achieved by NetFull and NetConv. This is likely since NetLin only performs a single linear transformation, limiting its capability in capturing more-complex, non-linear spatial features. Thus, the model is likely underfitting our data.

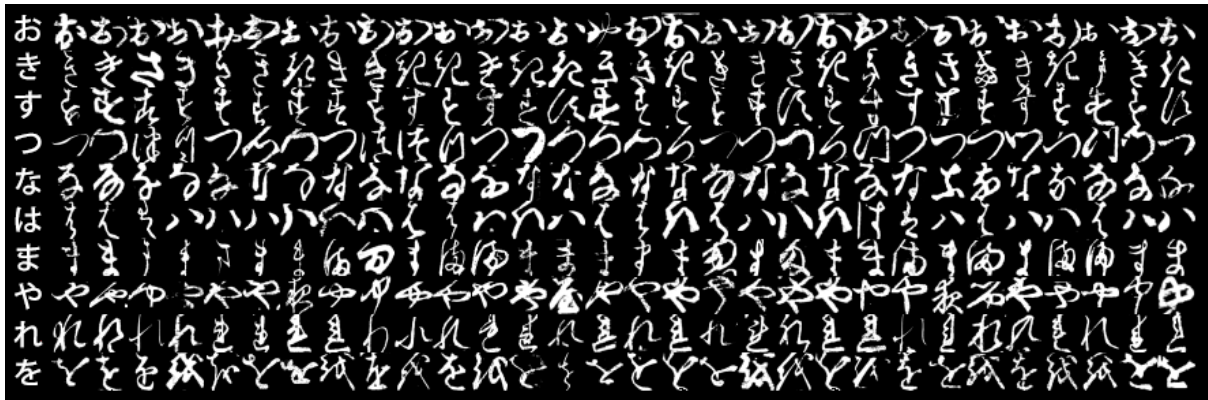
NetFull extends upon the limitations of NetLin, implementing a fully connected hidden layer (with 150 nodes) and a non-linear tanh activation function. It observes a significant improvement in performance, increasing from the 70% accuracy achieved by NetLin to an accuracy of 85%. However, NetFull is still limited in its inability to capture spatial features, which are not preserved after performing the flattening operation.

NetConv further extends upon the limitations of NetFull (for this task). Its use of convolution layers preserves the spatial features and relationships in the image data. Due to this, it can capture local features such as edges, curves and shapes. This results in a better fit and thus more distinctive separation when considering the decision boundaries. Therefore, we observe a significantly higher performance with NetConv, increasing from the 85% accuracy achieved by NetFull to an accuracy of 94%.

Number of Independent Parameters of Each Model

When simultaneously observing the number of parameters and accuracy of NetFull and NetConv, we see that despite using a fewer number of parameters, NetConv achieves a significantly higher accuracy. This can primarily be attributed to its architectural superiority for the required classification task. Specifically, the use of convolution layers which capture spatial feature, in combination with their weight-sharing property means that parameters are more efficiently used and used on features with more predictive power. Finally, NetLin's relatively low parameter count reinforces the notion that it may be underfitting for the required classification task.

Confusion Matrix Analysis for Each Model



*Above is some sample data from the KMNIST dataset, with the i^{th} row representing the i^{th} character. (0="o", 1="ki", 2="su", 3="tsu", 4="na", 5="ha", 6="ma", 7="ya", 8="re", 9="wo"). Further, each column represents a separate sample of the i^{th} character

Considering the confusion matrix of NetLin, we see that there is significant misclassification across all characters. Confusion is specifically higher on character 7="ya", with a misclassification rate of 37.5%. The character showcases a large variance in rotational, translation and pixel intensity. This likely results in an inability to learn a linear relationship between the input pixels and the class label. Thus, we observe a relatively high level of confusion with most other labels.

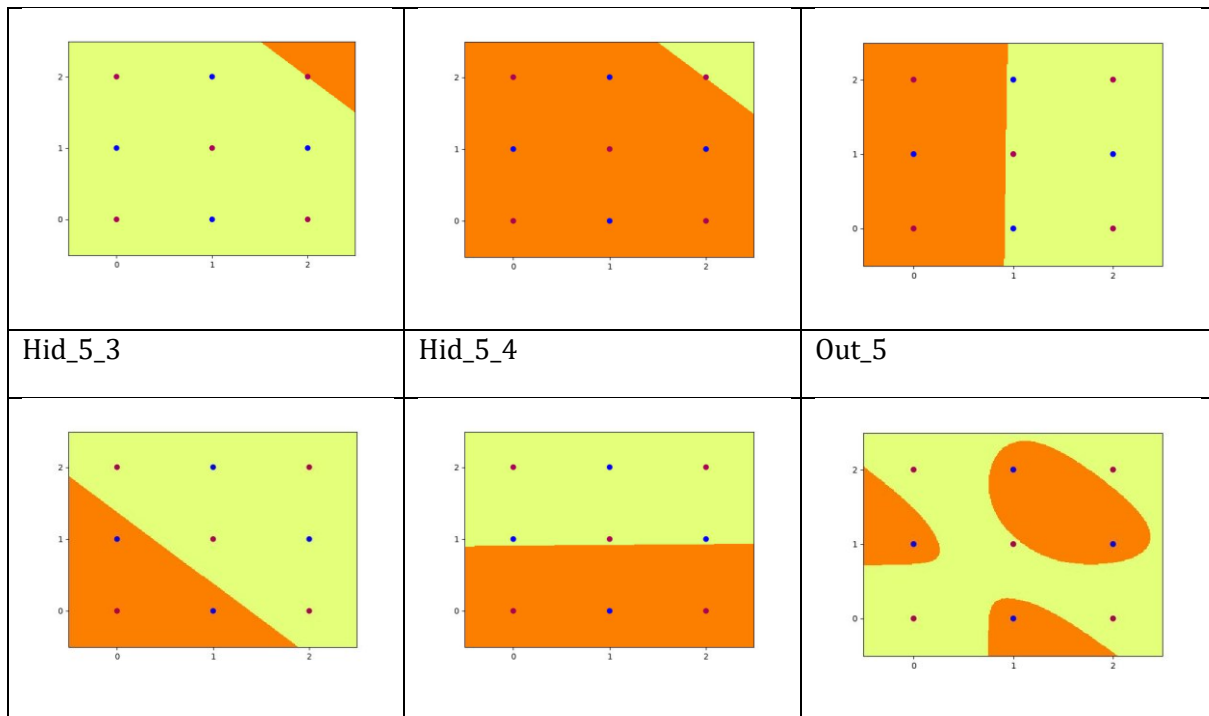
NetFull improves upon this due to its ability to model more complex relationships between input pixels and output labels, however the misclassification is still relatively significant. Under NetFull, character 1="ya" and 6="ma" observe one of the highest confusion (with "ya" predicted as "ma" 6% of the time). This is likely due to their similarity in strokes and general pixel intensity. Without the ability to identify spatial relationships in the data, the model is likely unable to separate the data on localised relationships.

The use of convolution layers in NetConv significantly improves upon this issue, reducing the misclassification of "ya" as "ma" from 6% to 3.8%. The overall confusion is significantly reduced, suggesting that the ability to capture spatial features allows the model to draw more distinctive decision boundaries between classes. However, there remains a significant misclassification between characters 2="su" as 3="tsu". This is likely due to the similar representations of both characters across different handwriting styles; however, it outlines the clear imperfections of the CNN. This reveals the opportunity for improvement by considering alternative features such as context.

Part 2: Multi-Layer Perceptron

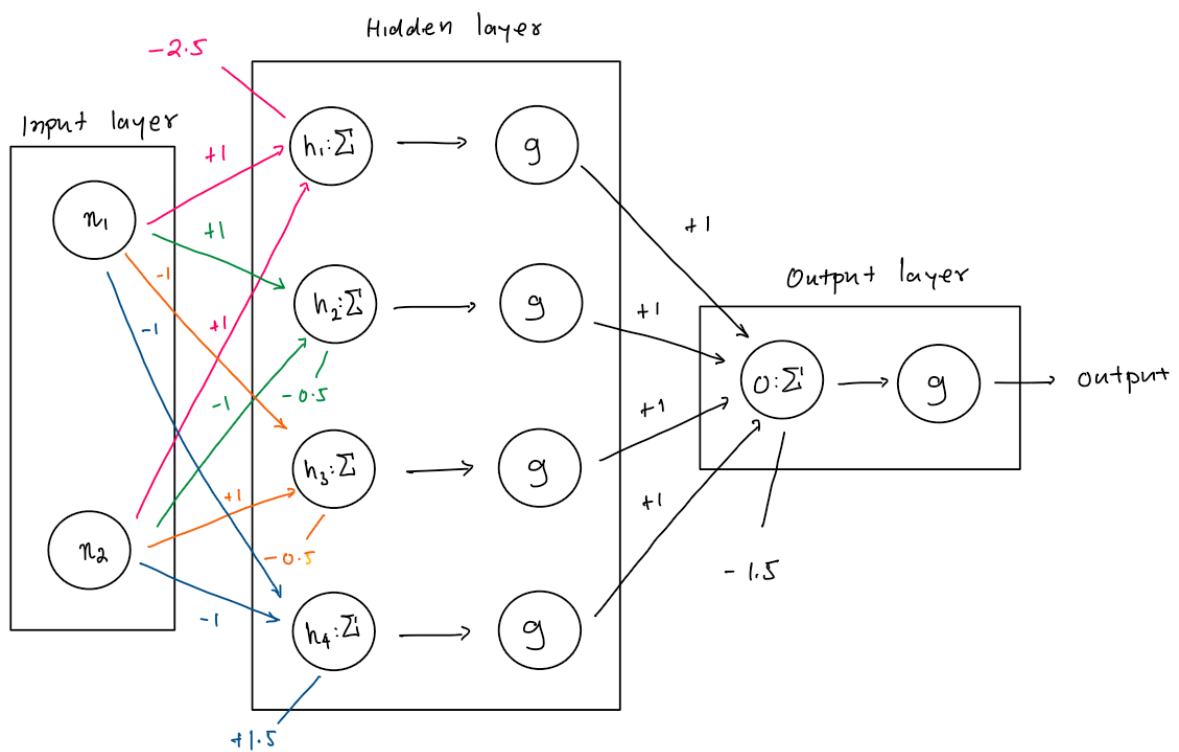
Question 1

Hid_5_0	Hid_5_1	Hid_5_2
---------	---------	---------

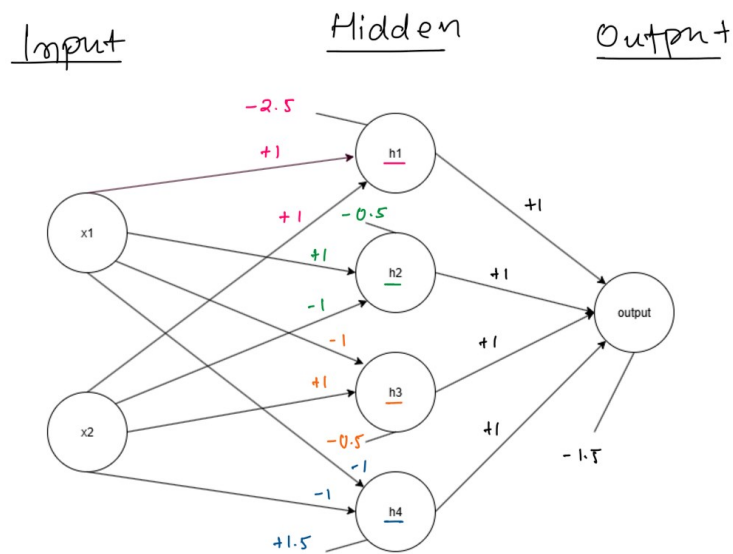


Question 2

Network Diagram



Here, g represents the Heaviside activation function. Specifically, $g(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$. This can be omitted from the network diagram to obtain the following simpler diagram.



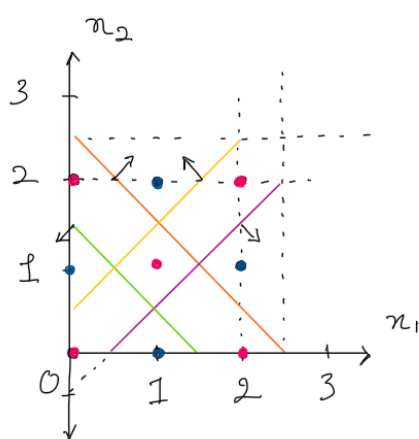
Network Weights (Clarification)

```

in_hid_weight = [[1,1],[1,-1],[-1,1],[-1,-1]]
hid_bias      = [-2.5,-0.5,-0.5,1.5]
hid_out_weight = [[1,1,1,1]]
out_bias      = [-1.5]

```

Hidden Node Equations



$$\begin{aligned}
 +H_1 &\geq 0 & +H_3 &\geq 0 \\
 -H_2 &\geq 0 & -H_4 &\geq 0
 \end{aligned}$$

class 0
class 1

$$H_1: n_2 = \frac{5}{2} - n_1 \Rightarrow n_1 + n_2 - \frac{5}{2} \geq 0$$

$$H_2: n_2 = -\frac{1}{2} + n_1 \Rightarrow n_1 - n_2 - \frac{1}{2} \geq 0$$

$$H_3: n_2 = \frac{1}{2} + n_1 \Rightarrow -n_1 + n_2 - \frac{1}{2} \geq 0$$

$$H_4: n_2 = \frac{3}{2} - n_1 \Rightarrow -n_1 - n_2 + \frac{3}{2} \geq 0$$

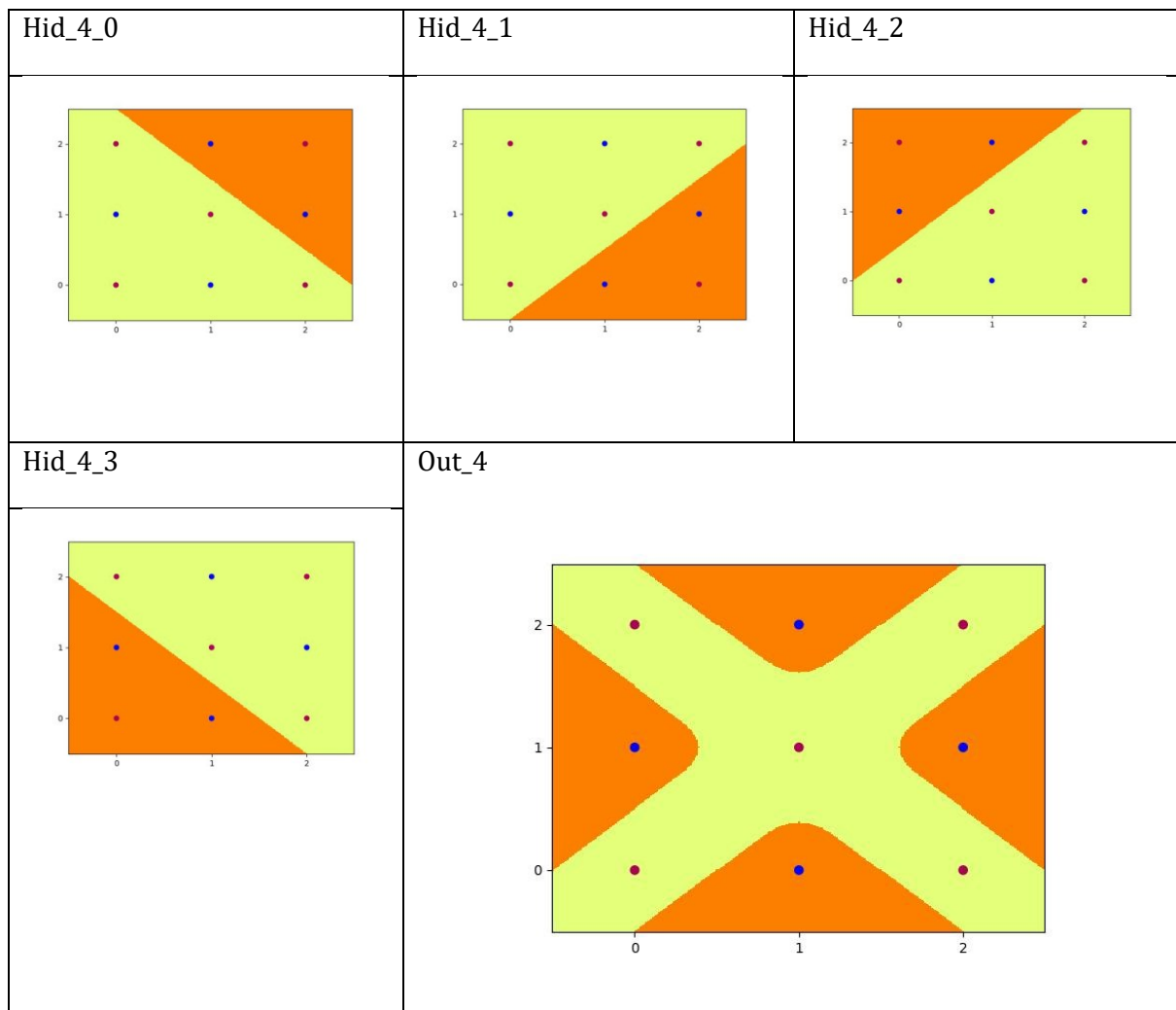
Activation of Hidden & Output Nodes

X	Y	T	z1	z2	z3	z4	g(z1)	g(z2)	g(z3)	g(z4)	O1	g(O1)	Correct?
0	0	0	-2.5	-0.5	-0.5	1.5	0	0	0	1	-0.5	0	TRUE
0	1	1	-1.5	-1.5	0.5	0.5	0	0	1	1	0.5	1	TRUE
0	2	0	-0.5	-2.5	1.5	-0.5	0	0	1	0	-0.5	0	TRUE
1	0	1	-1.5	0.5	-1.5	0.5	0	1	0	1	0.5	1	TRUE
1	1	0	-0.5	-0.5	-0.5	-0.5	0	0	0	0	-1.5	0	TRUE
1	2	1	0.5	-1.5	0.5	-1.5	1	0	1	0	0.5	1	TRUE
2	0	0	-0.5	1.5	-2.5	-0.5	0	1	0	0	-0.5	0	TRUE
2	1	1	0.5	0.5	-1.5	-1.5	1	1	0	0	0.5	1	TRUE
2	2	0	1.5	-0.5	-0.5	-2.5	1	0	0	0	-0.5	0	TRUE

Notation:

- 1) The columns labelled z1, z2, z3 and z4 are the linear combinations of hidden layer inputs and weights for hidden layers 1, 2, 3 and 4 respectively.
- 2) g is the Heaviside activation function. Here $g(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$.
- 3) $g(z1), g(z2), g(z3), g(z4)$ are the outputs for hidden layers 1, 2, 3 and 4.
- 4) O1 refers to the linear combinations of output layer inputs and weights. $g(O1)$ is the output of the output node. Once again, g refers to the Heaviside activation function.

Question 3



NOTE, the initial weights were multiplied by 10. This resulted in the following set of weights:

```

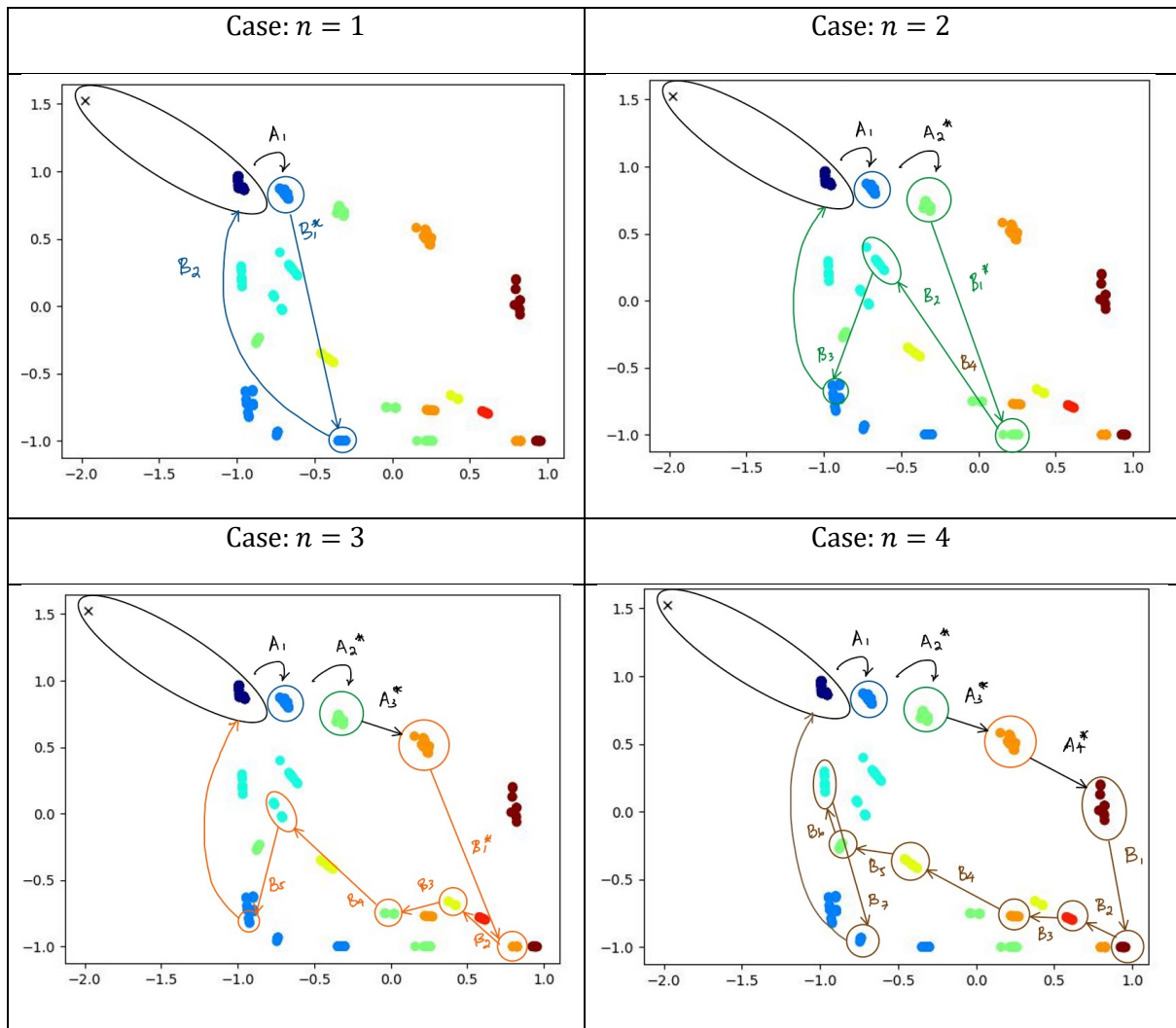
in_hid_weight  = [[10,10],[10,-10],[-10,10],[-10,-10]]
hid_bias       = [-25,-5,-5,15]
hid_out_weight = [[10,10,10,10]]
out_bias       = [-15]
  
```

Part 3: Hidden Unit Dynamics for Recurrent Networks

Question 1

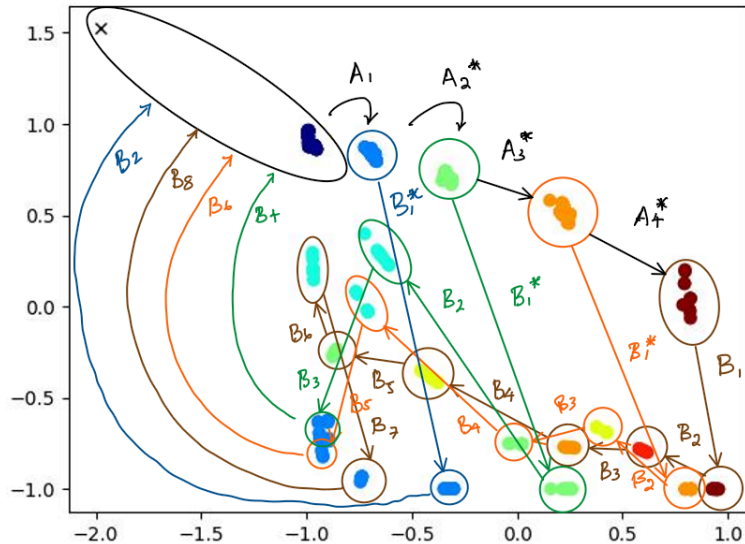
Trajectory Plots

When plotting the trajectories for each case ($n = 1, 2, 3, 4$) individually, we obtain the following diagrams:



Note that the * superscript denotes transitions which are non-deterministic. Further, the $\times$ denotes the initial state, and the dark blue cluster denotes the start of a new sequence. These have been grouped into a single state as suggested by Dr Alan Blair on the course forum.

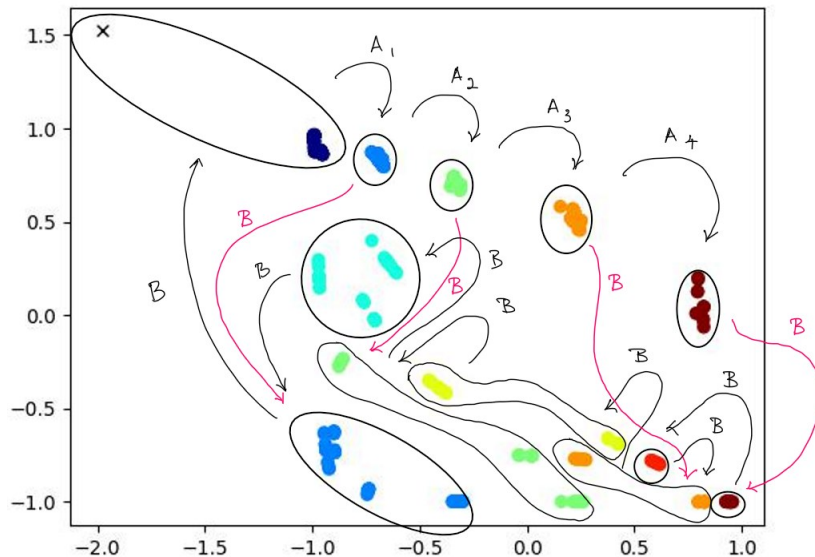
After combining these plots, we get the final trajectories/ states:



Note that whilst some slight changes have been made for the sake of neatness/ readability, the plots are logically equivalent.

Justification for Selected States

A key question can be raised when creating the plots above – do you group sets of points together that are extremely close in proximity and exhibit similar behaviour. For example, should the blue points circled in green, and orange be grouped together? Applying such principles may result in separate plots such as the following:

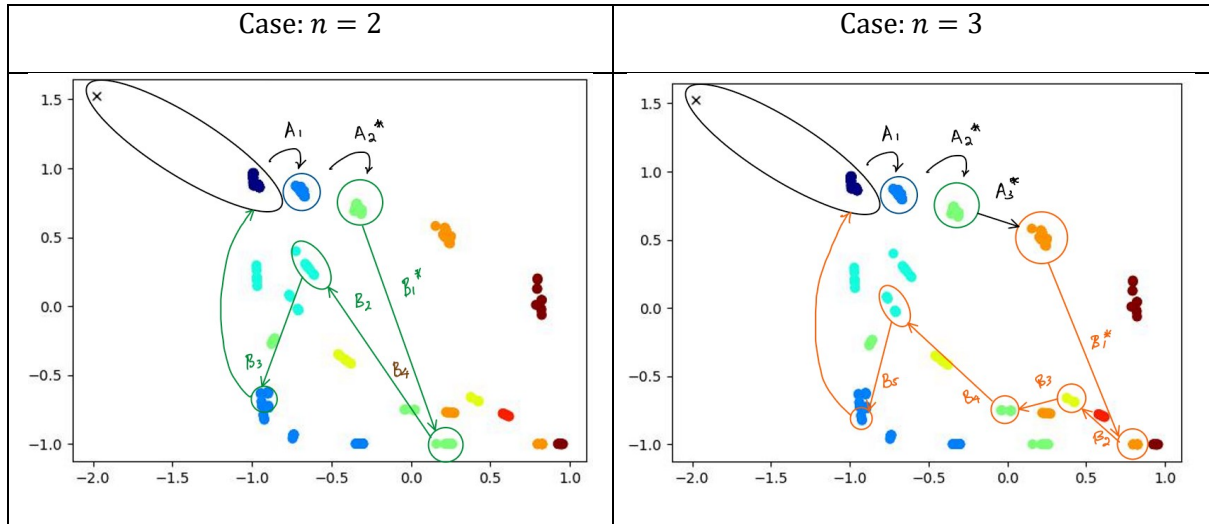


My conclusions when investigating this was that they **should not be grouped**. Consider the justification below.

Hidden states are updated according to:

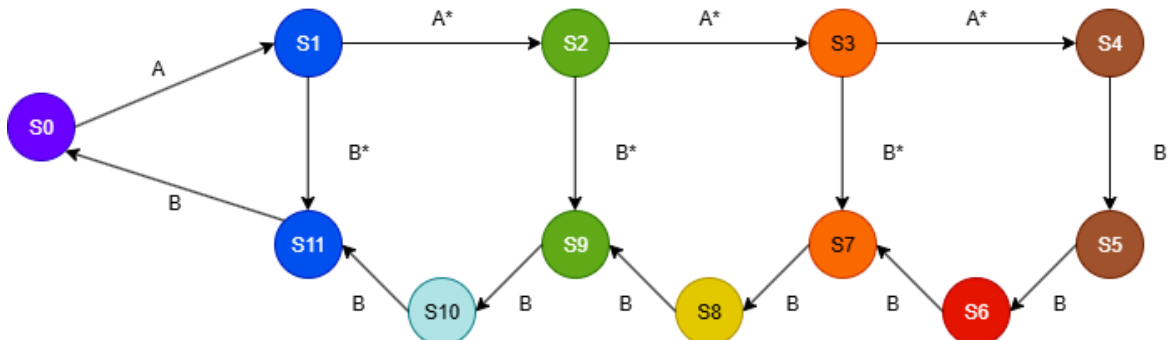
$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b)$$

This means that as h is recursively updated, it accumulates information on $h_0, \dots, h_{t-1}$.



Thus, when considering the example discussed above, we see that whilst the states around the blue points are extremely close in proximity and “deterministically” transition to the initial state with the annotation B_{n-1} , they are inherently different since each of their h_t contain different information. This is because they follow different trajectories to reach their respective states.

Question 2



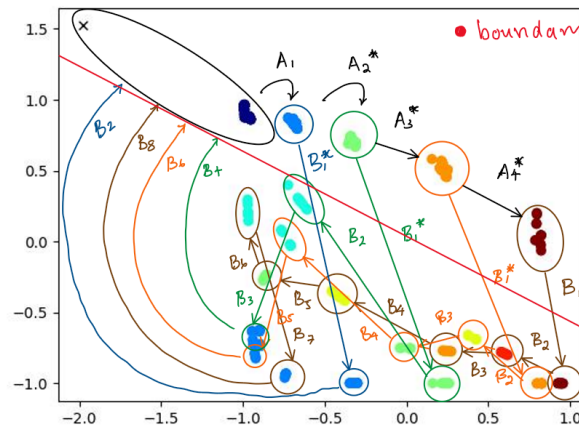
Once again, the * superscript denotes transitions which are non-deterministic

Question 3

The RNN can model the $a^n b^{2n}$ problem by implicitly learning to count the number of characters A. It does this by learning the trajectories (transitions) and hidden states associated with different stages (t) in a particular sequence of characters. As the RNN reads the character ‘A’ from its input, it recursively updates its hidden state using:

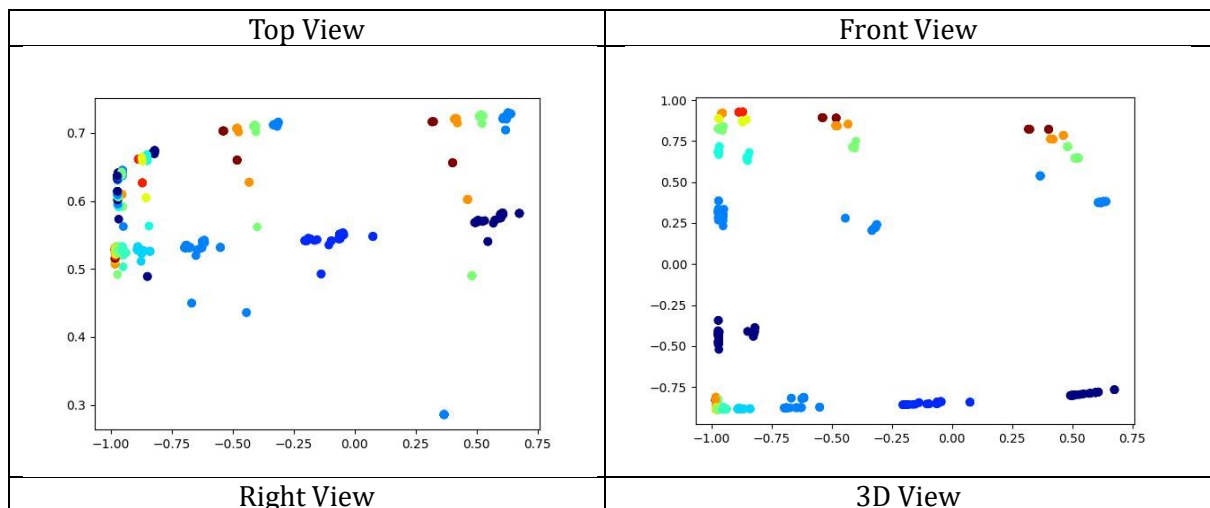
$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b)$$

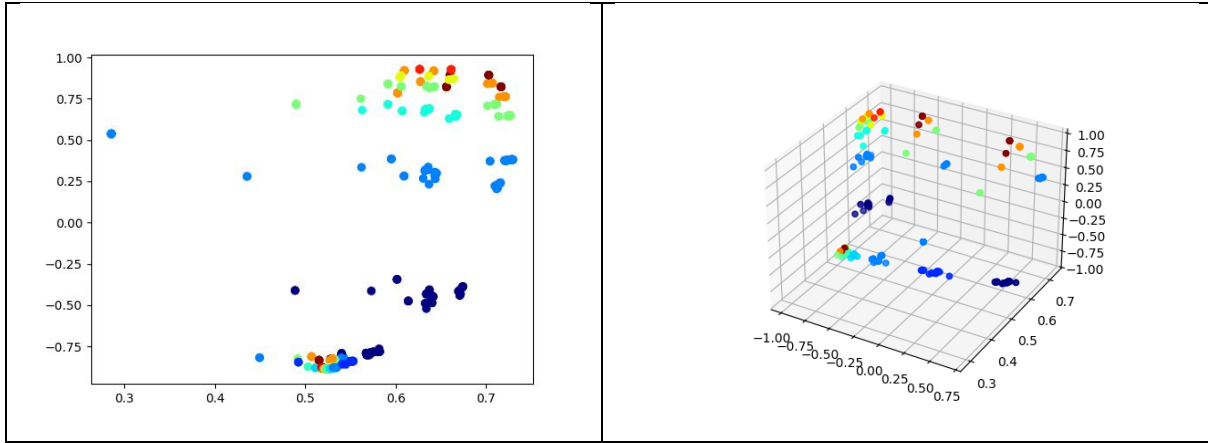
It is important to note that h_t contains information on $h_0, \dots, h_{t-1}$. Thus, using this, the network can map the count of ‘A’ to a vector on the hidden space (Rodriguez et al. (1999)).



When the model reads the first 'B' from its inputs, the updates for h_t showcase a sharp change in location over hidden space. This is analogous to understanding that you must shift from incrementing to decrementing. This is reflected by the clear distinction between the location of states that transition using 'A' and states that transition using 'B' (the red boundary in the image above). In this lower region, transitions using 'B' are deterministic given the "saddle point" (Rodriguez et al. (1999)). The saddle point is the location of the first state after transitioning to the 'B' region, which decides the (remaining) deterministic trajectory of the sequence. This means that if the model thinks it had read n 'A', then it will likely deterministically follow a deterministic trajectory following from the n^{th} 'A' to the first 'B' to the input state. Similarly, after the final 'B', the model reaches the initial state from which its trajectory deterministically output an A to begin the next sequence.

Question 4





Question 5

The plots in the following sections were generated using data that was printed using a modified version of the source code. The outputs of this modified code are provided in the appendix. It is important to note that the dynamics plotted were using epoch 4 as it contained all cases of the $a^n b^{2n} c^{3n}$ problem. That is, for cases $n = 1, 2, 3$ and 4.

Required Task & Limitations of RNN

Like the $a^n b^{2n}$ problem, $a^n b^{2n} c^{3n}$ requires the model to implicitly count the number of 'A' (increment), then decrement $2n$ 'B', then finally $3n$ 'C'. This problem involves additional complexity compared to $a^n b^{2n}$, specifically requiring the chosen model to "remember" the value of n over more timesteps. The RNN struggles at modelling the problem given this additional complexity since it aims to store information regarding the value of n , decrement count, the corresponding hidden state and transition (trajectory). Recall, $h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b)$. Thus, over larger sequences, the RNN is very likely to suffer from instability in values of h_t since h_{t-1} (information regarding the past states) is being partially over-written by x_t .

Solutions using the LSTM Architecture:

The LSTM's hidden dynamics behave differently compared to the RNN. They are dependent on:

**Note, the following definitions are taken from the COMP9444 25T2 RNN Lecture Slides*

- 1) Forget Gate - what information to remove from the cell state:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

- 2) Candidate State - what to potentially write into memory:

$$\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

- 3) Input Gate – how much of the candidate state should be written to memory:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

- 4) Output Gate – how much of the current memory goes to output:

$$o_t = \sigma(W_o x_t + U_o h_{t-1})$$

These then combine to form the Cell State (C_t often referred to as context), which is used to calculate the hidden activation h_t :

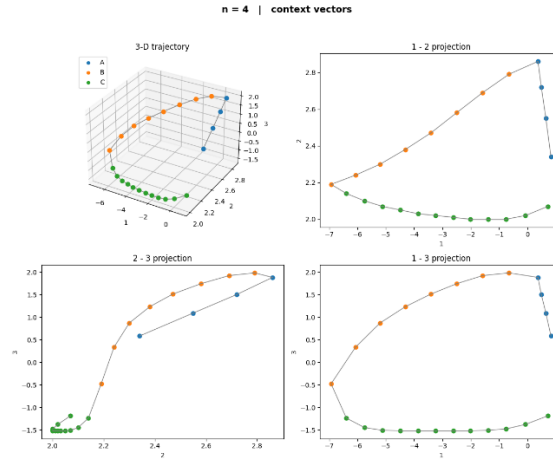
$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t$$

$$h_t = o_t \times \tanh(C_t)$$

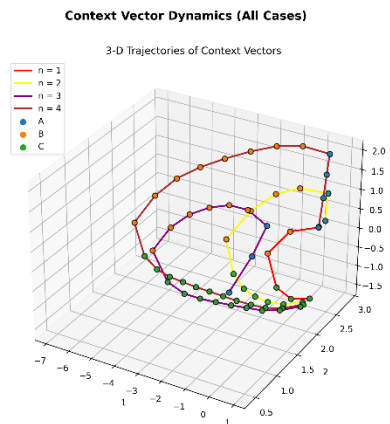
Unlike the RNN which over-writes its value of h_t , the LSTM's context is updated in an additive manner. These updates are also "gated," meaning the model has much more control over how inputs and past context contribute towards the updated cell state. In combination, these make the LSTM's memory much more "stable" and "persistent".

Analysis of LSTM Context Dynamics

Below is a plot showcasing the dynamics of the LSTM's context. As the LSTM reads the character 'A' from input, we observe that successive updates are relatively small and follow a well-defined trajectory. At this stage, the LSTM is implicitly "counting" the value of n and thus, its update $C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t$ would likely yield values of f_t and i_t closer to 1. During this, the candidate state $\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$ is capturing information regarding the current inputs ('A') and previous hidden states. These are accumulated on top of the previous context calculated by the network. We observe that the context consistently increases across dimensions 2 and 3 in a linear manner but observes a slight decrease in dimension 1.

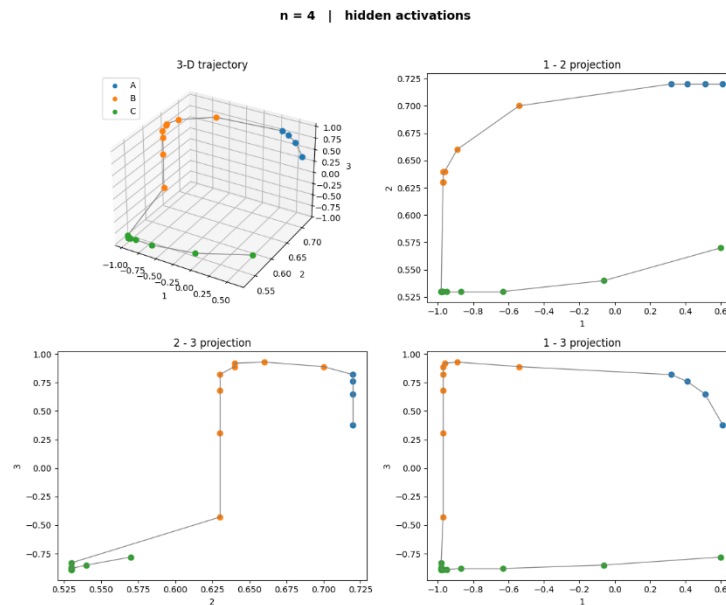


Like the RNN, when the LSTM reads the first 'B' from its inputs, we observe a sharp shift in the dynamics of the context. This shift is driven by a change in the behaviour of the cell state which is once again analogous to understanding that you must shift from incrementing to decrementing. We see that the context begins to decrease over all dimensions, however like previously, this decrease is relatively small and follow a well-defined trajectory. After the final 'B', a similar pattern is observed in the trajectory of the model's context as it begins to process $3n$ 'C' characters. That is, the context sees sharp shift in direction wherein axis 1 increases significantly, axis 2 decreases slight, and axis 3 remains roughly the same, with each changing in a stable but gradual manner. These trajectories are observed under all cases $n = 1, 2, 3, 4$, as shown in the plot below.



Analysis of LSTM Hidden Activation Dynamics

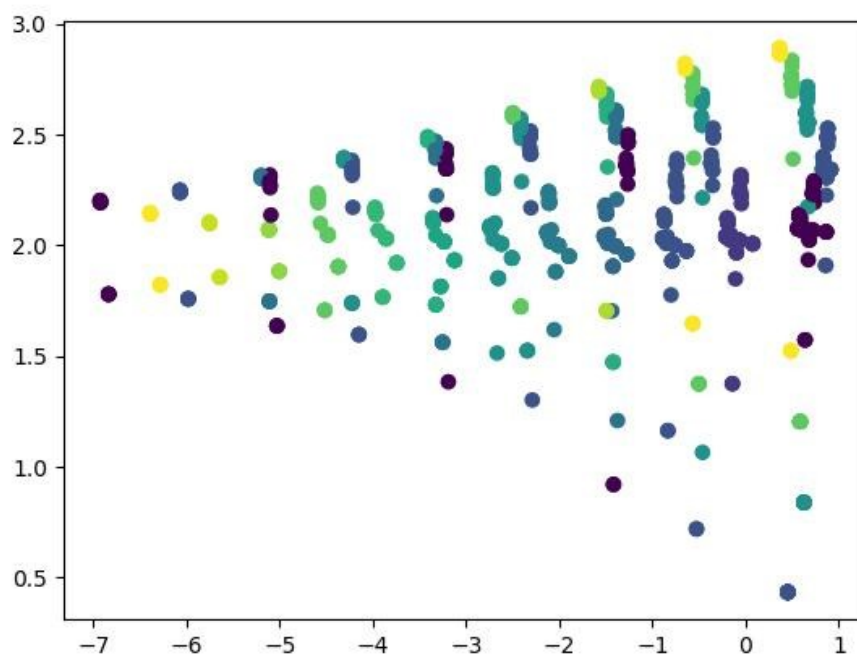
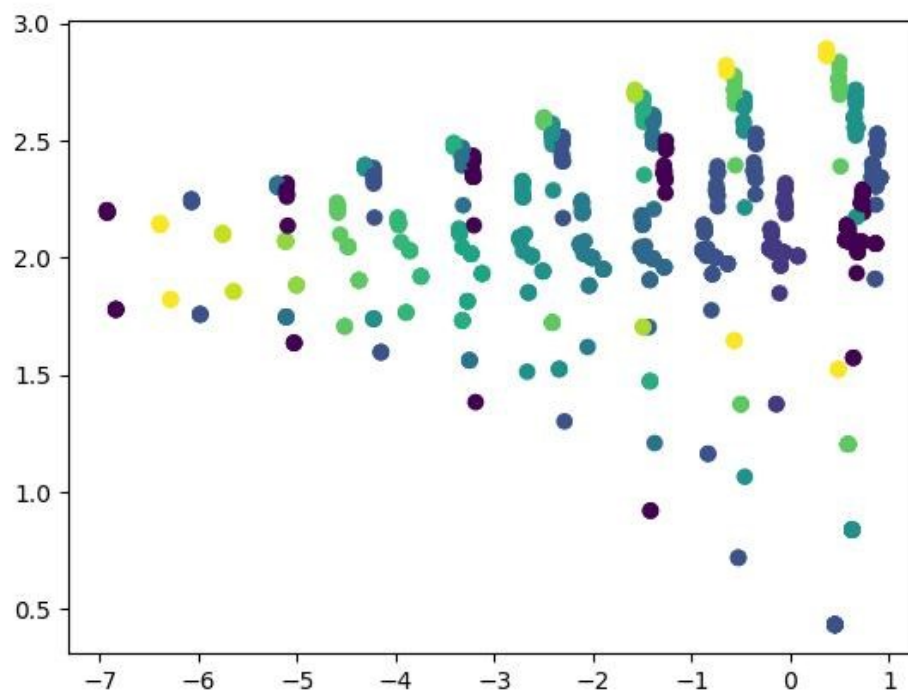
The trajectories of the hidden activation for the LSTM behave similarly to the RNN. That is, they are deterministic from the first 'B' in a sequence to the first 'A' in the next sequence. These trajectories are once again characterised by a “saddle” point, which we can observe in the plot below. Further, there is a clear distinction between the location of states that transition using 'A', 'B' and 'C', and a sharp change in dynamics when these transitions occur.

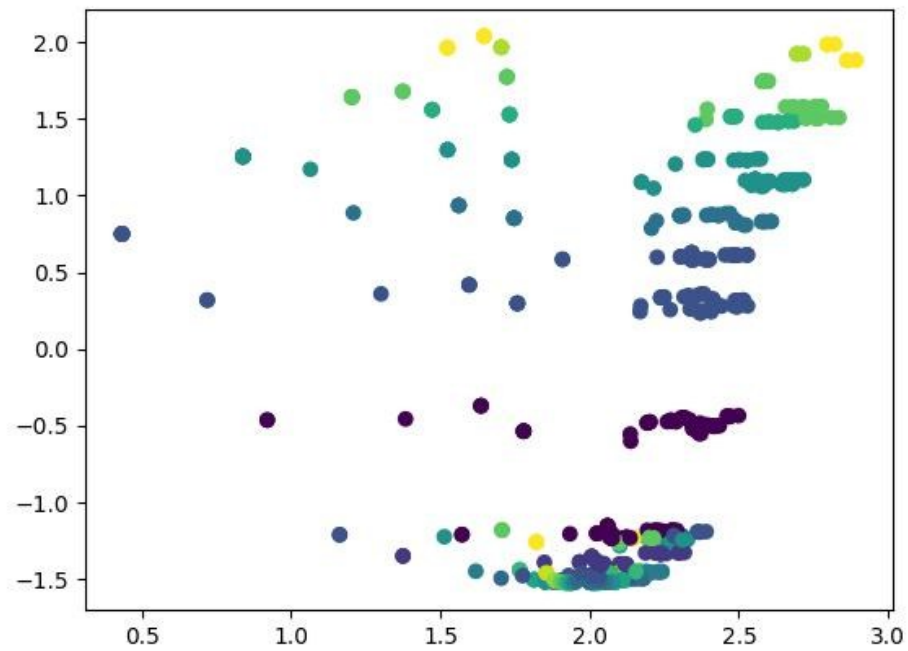


However, recall that under the LSTM model, $h_t = o_t \times \tanh(C_t)$. The key differences lie in the LSTM's use of context units, which allow it to learn dependencies in the $a^n b^{2n} c^{3n}$ problem over a larger number of steps.

Appendix

Context Vector Plots for P3Q5



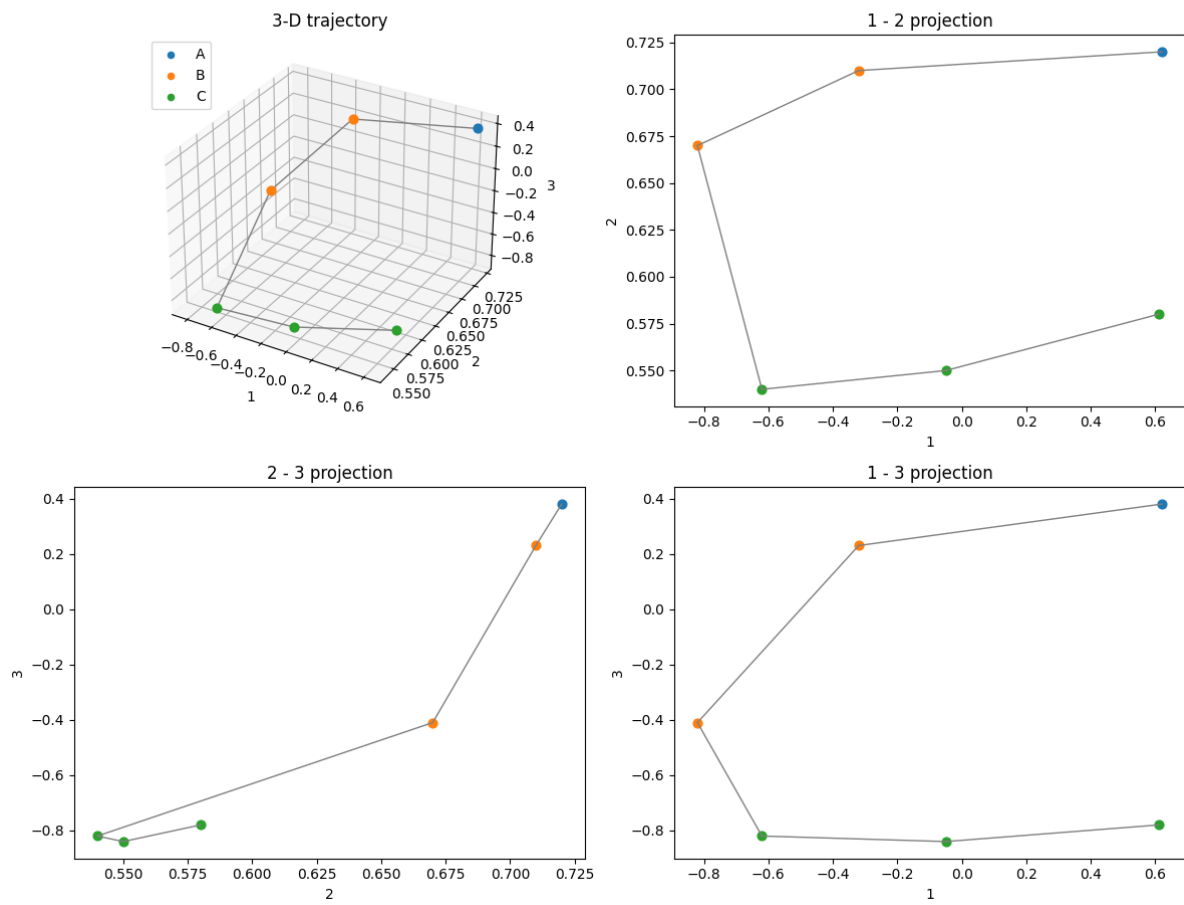


Hidden and Context Dynamics for P3Q5

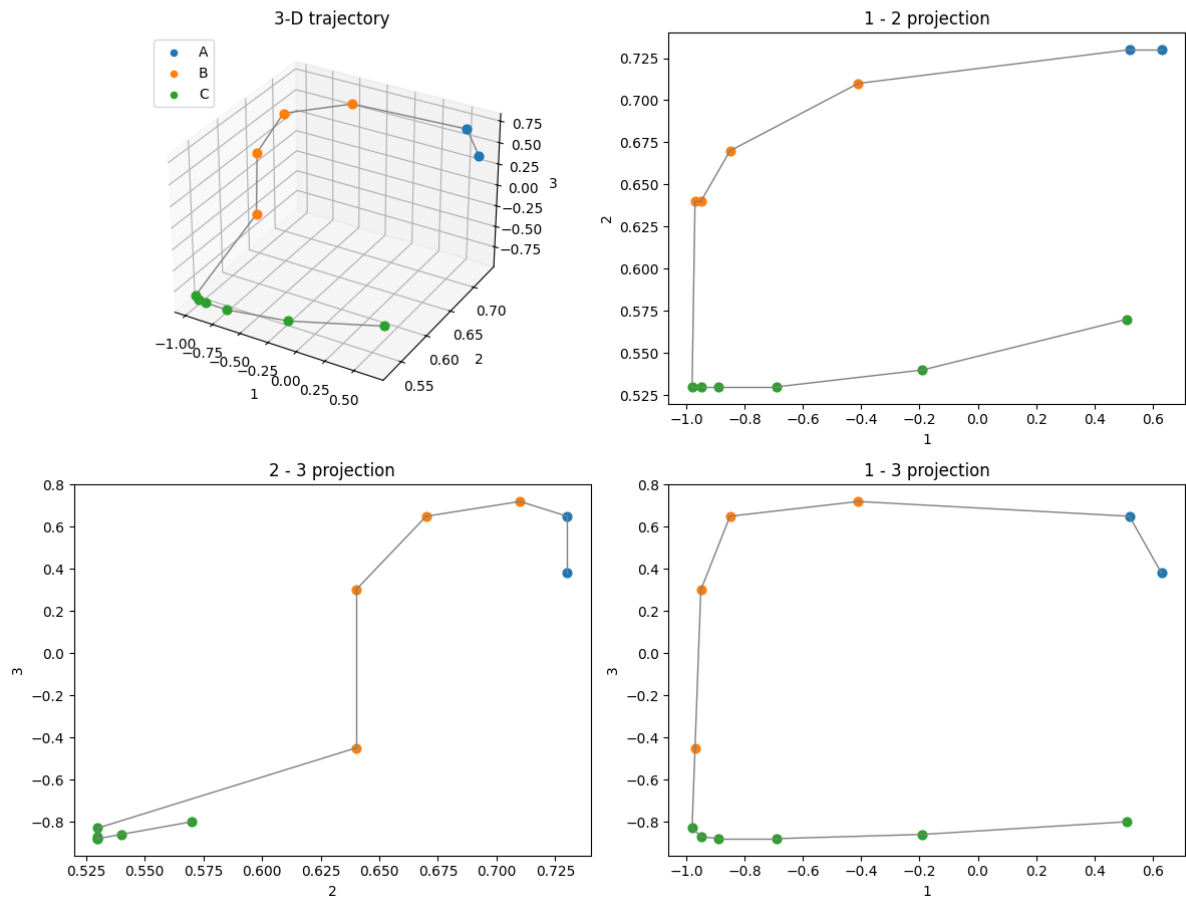
Note, these were generated using the data from epoch 4. All values are listed in the appendix section titles 'All Hidden Activations, Probs & Context Vectors for P3Q5'.

Hidden Dynamics

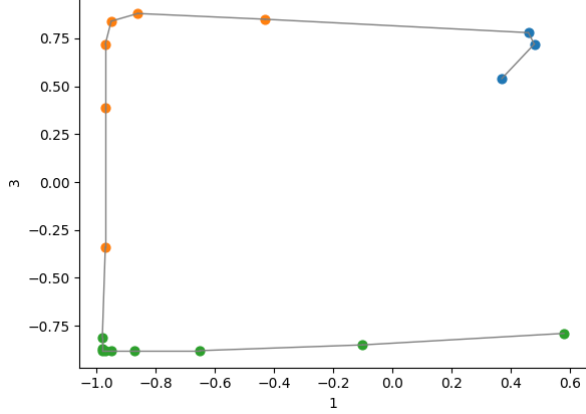
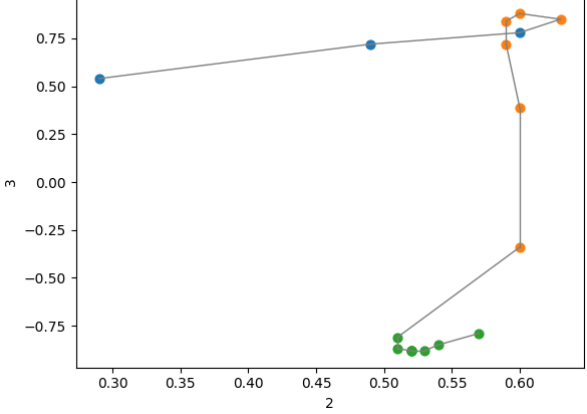
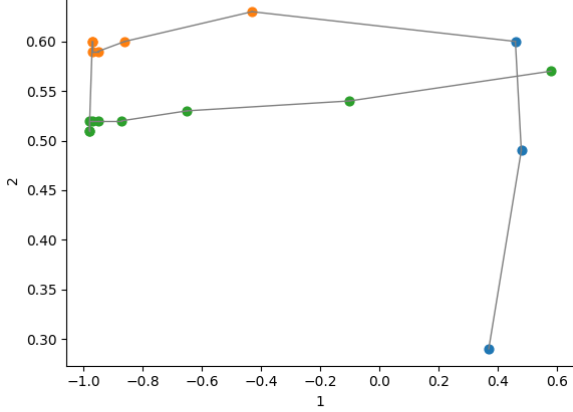
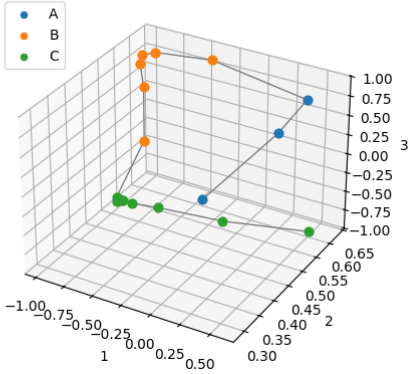
n = 1 | hidden activations



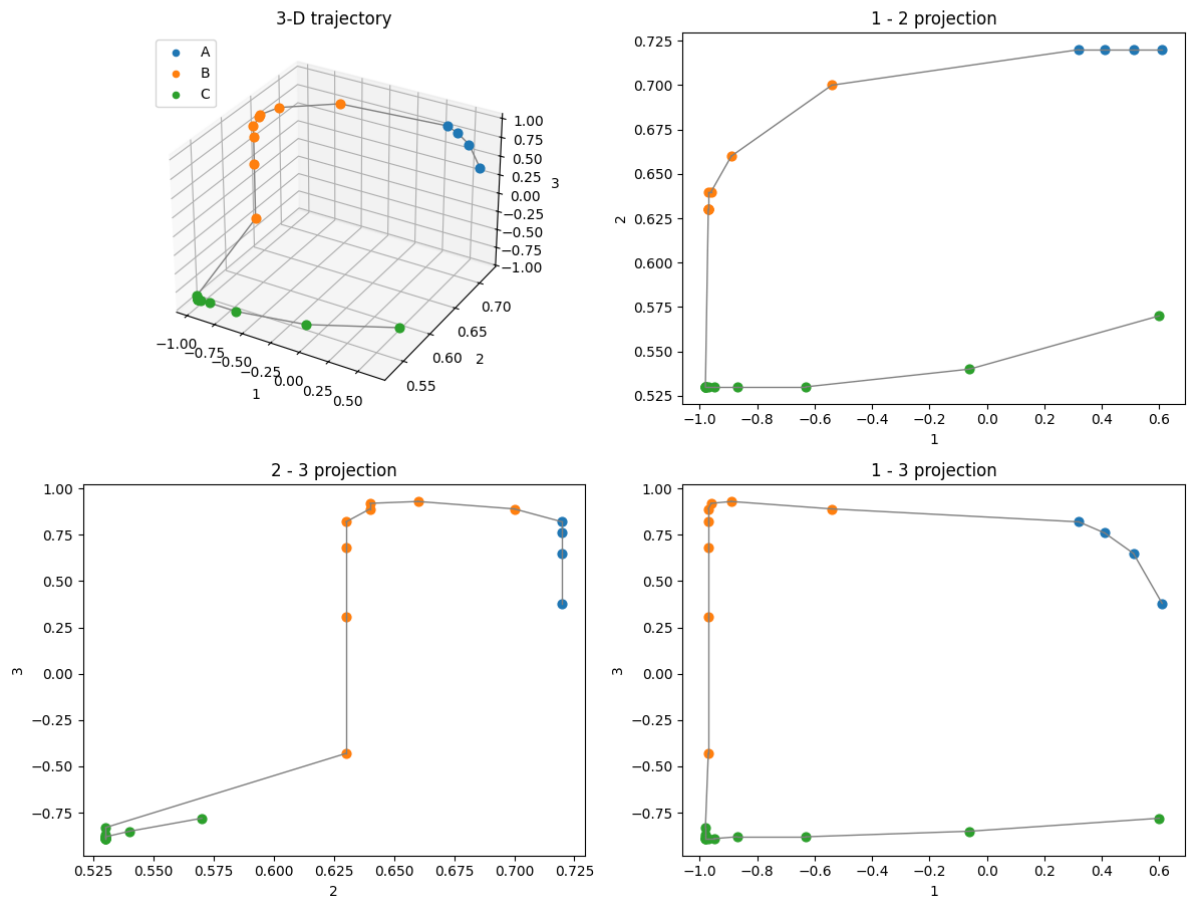
n = 2 | hidden activations



n = 3 | hidden activations

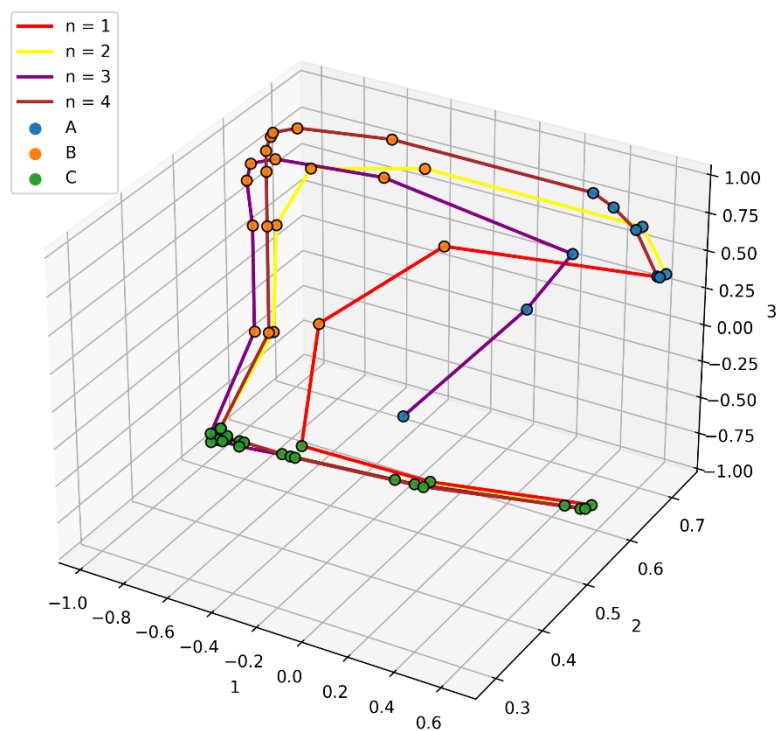


n = 4 | hidden activations



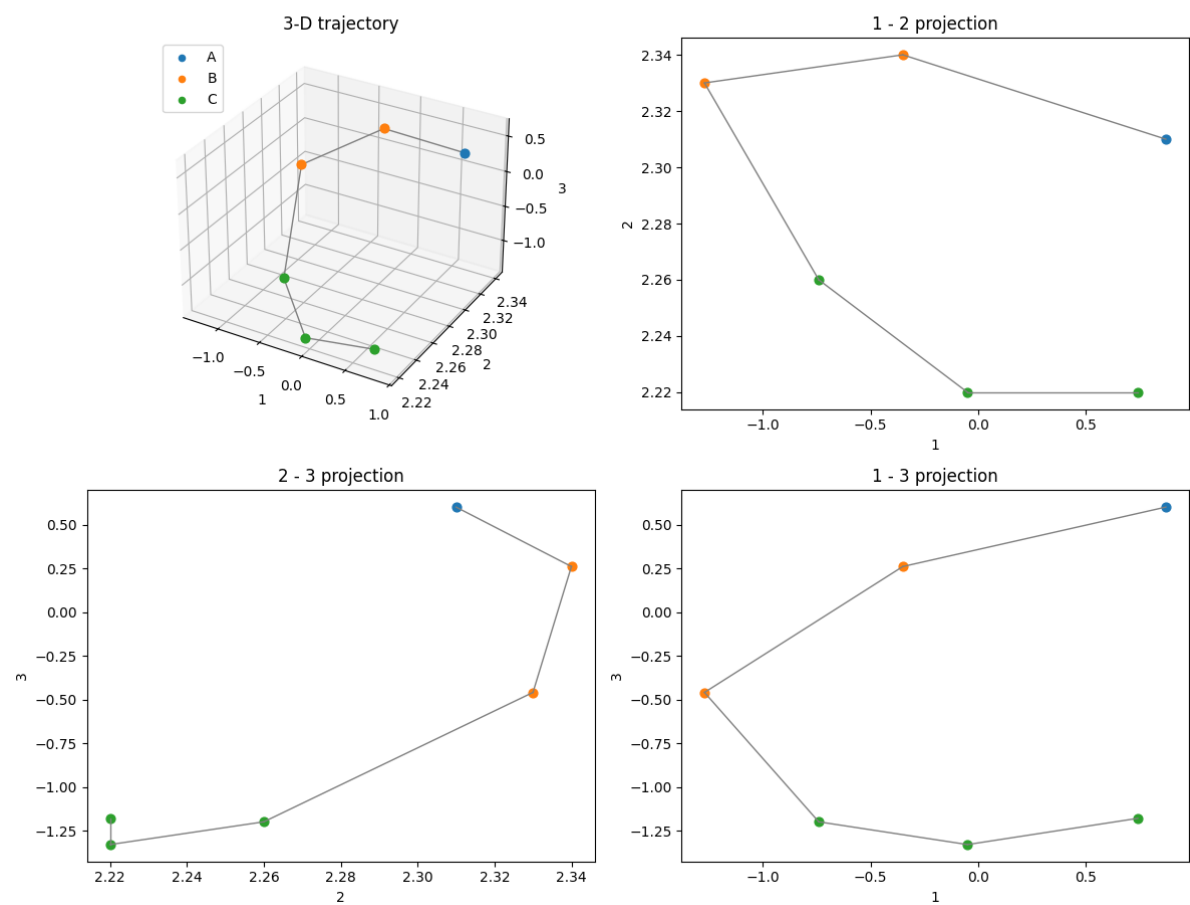
Hidden Activation Dynamics (All Cases)

3-D Trajectories of Hidden States

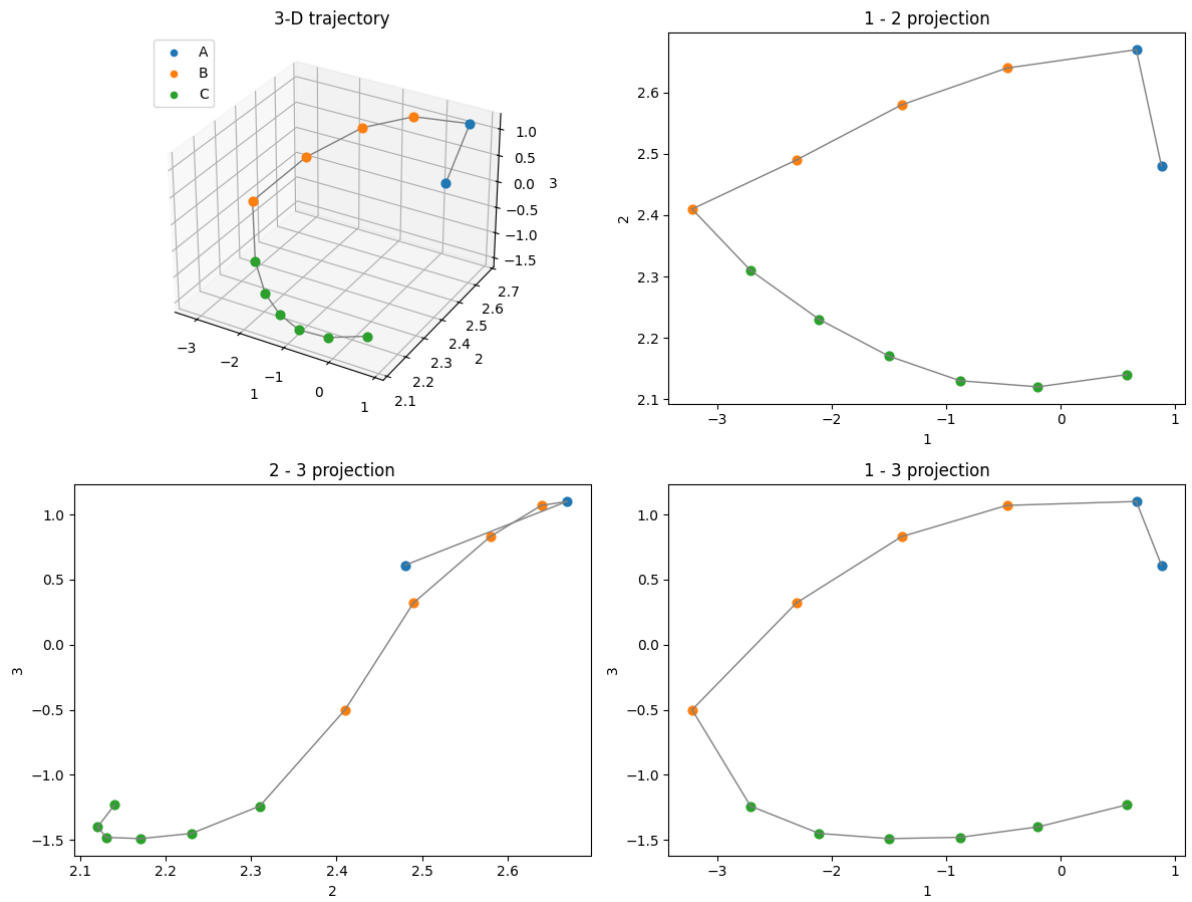


Context Dynamics

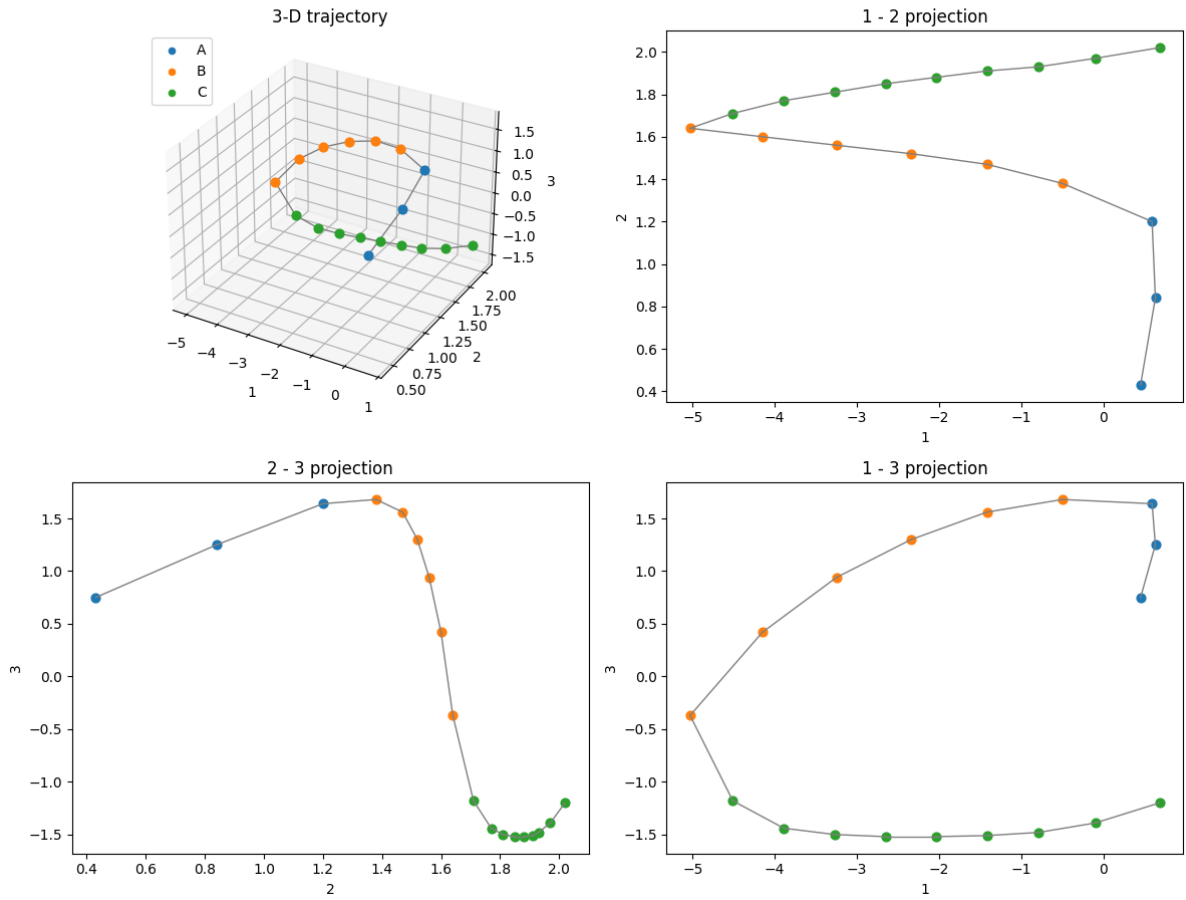
n = 1 | context vectors



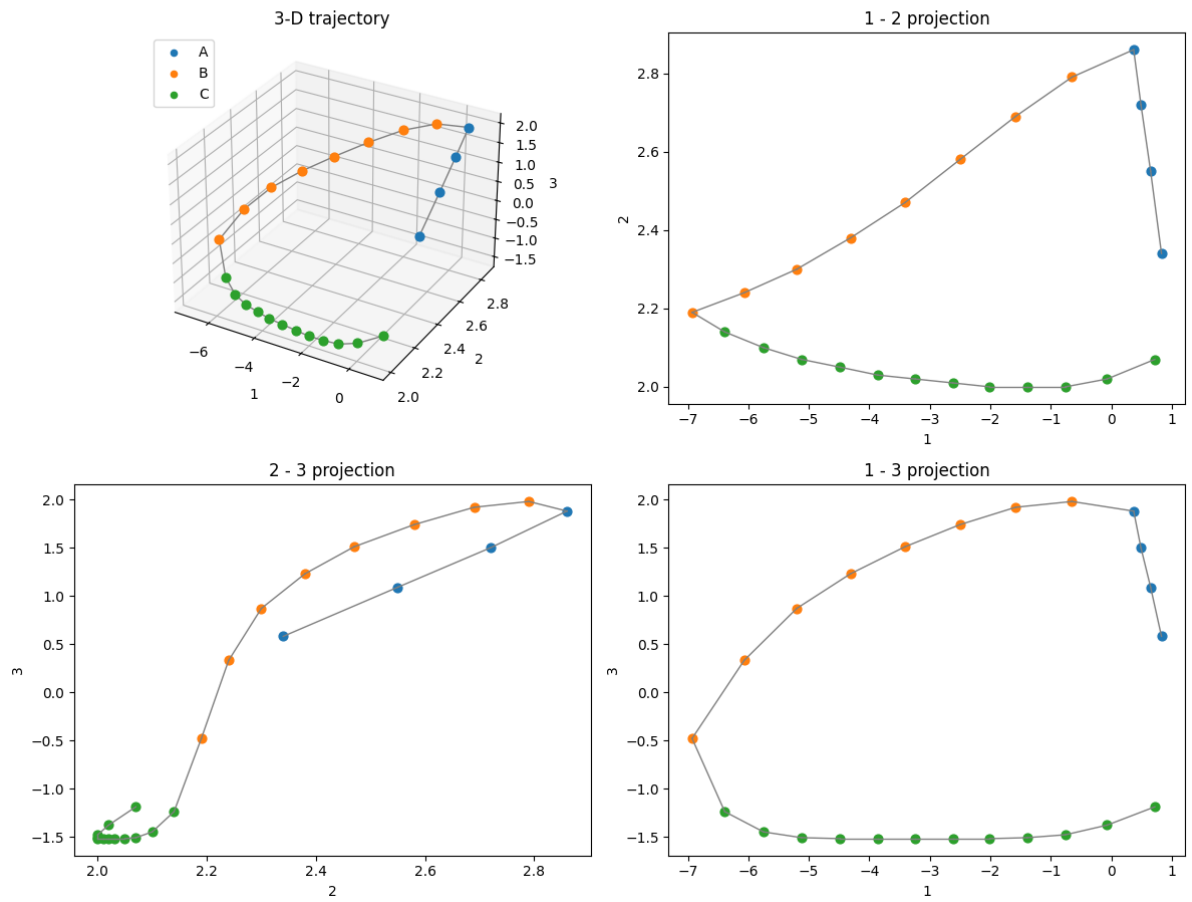
n = 2 | context vectors



n = 3 | context vectors

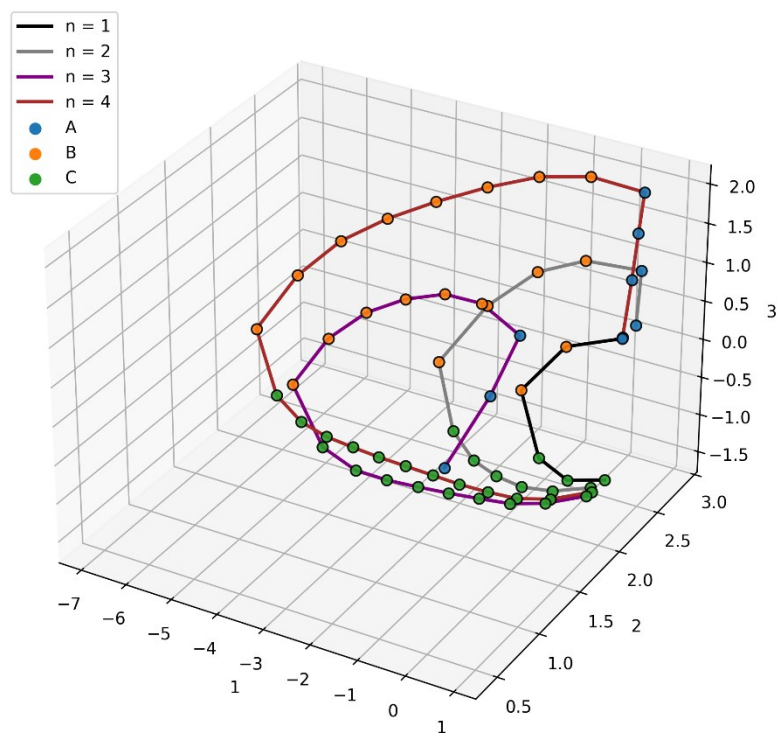


n = 4 | context vectors



Context Vector Dynamics (All Cases)

3-D Trajectories of Context Vectors



All Hidden Activations, Probs & Context Vectors for P3Q5

(see next 10 pages)

```
-----
symbol= AAAABBBBBBBBBCCCCCCCCCCCCAABBBBBCCCCCAABBBBBCCCCCAAAABBBBBBBBBCCCCCCCCCCCCAABBBBBCCCCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
A [0.48 0.49 0.72] [0.51 0.49 0. ] [0.63 0.84 1.25]
A [0.46 0.6 0.78] [0.4 0.6 0. ] [0.59 1.2 1.64]
A [0.4 0.66 0.82] [0.29 0.71 0. ] [0.49 1.52 1.96]
B [-0.48 0.66 0.89] [0. 1. 0.] [-0.57 1.65 2.04]
B [-0.87 0.63 0.93] [0. 1. 0.] [-1.49 1.7 1.97]
B [-0.96 0.61 0.92] [0. 1. 0.] [-2.41 1.72 1.77]
B [-0.97 0.61 0.89] [0. 1. 0.] [-3.32 1.73 1.53]
B [-0.97 0.61 0.82] [0. 1. 0.] [-4.22 1.74 1.23]
B [-0.97 0.61 0.68] [0. 0.99 0.01] [-5.11 1.75 0.85]
B [-0.97 0.61 0.28] [0. 0.9 0.09] [-5.98 1.76 0.3 ]
B [-0.97 0.61 -0.47] [0. 0.04 0.96] [-6.83 1.78 -0.54]
C [-0.98 0.52 -0.83] [0. 0. 1.] [-6.28 1.82 -1.26]
C [-0.98 0.52 -0.87] [0. 0. 1.] [-5.64 1.86 -1.46]
C [-0.98 0.52 -0.88] [0. 0. 1.] [-5. 1.88 -1.51]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-4.37 1.9 -1.52]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-3.74 1.92 -1.52]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-3.12 1.93 -1.52]
C [-0.97 0.52 -0.89] [0. 0. 1.] [-2.51 1.94 -1.52]
C [-0.94 0.52 -0.88] [0. 0. 1.] [-1.89 1.95 -1.52]
C [-0.84 0.53 -0.88] [0. 0. 1.] [-1.28 1.96 -1.5 ]
C [-0.55 0.53 -0.87] [0. 0. 0.99] [-0.64 1.97 -1.46]
C [ 0.07 0.55 -0.84] [0.23 0.01 0.76] [ 0.08 2.01 -1.35]
C [ 0.68 0.58 -0.77] [0.95 0. 0.05] [ 0.86 2.06 -1.15]
A [0.64 0.73 0.38] [0.85 0.14 0. ] [0.91 2.34 0.63]
A [0.53 0.73 0.65] [0.57 0.43 0. ] [0.67 2.55 1.11]
B [-0.41 0.71 0.72] [0.01 0.99 0. ] [-0.46 2.54 1.08]
B [-0.85 0.67 0.66] [0. 0.99 0.01] [-1.38 2.49 0.84]
B [-0.95 0.64 0.31] [0. 0.92 0.07] [-2.3 2.41 0.33]
B [-0.97 0.64 -0.44] [0. 0.05 0.95] [-3.21 2.35 -0.49]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.71 2.26 -1.24]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.19 -1.45]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.14 -1.49]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.87 2.11 -1.48]
C [-0.18 0.54 -0.86] [0.05 0.01 0.94] [-0.19 2.1 -1.4 ]
C [ 0.51 0.57 -0.8 ] [0.86 0. 0.14] [ 0.58 2.12 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.39 0.58]
A [0.51 0.72 0.65] [0.56 0.44 0. ] [0.65 2.59 1.09]
B [-0.41 0.71 0.71] [0.01 0.99 0. ] [-0.47 2.57 1.06]
B [-0.85 0.67 0.64] [0. 0.99 0.01] [-1.4 2.52 0.81]
B [-0.95 0.64 0.27] [0. 0.9 0.1] [-2.32 2.44 0.28]
B [-0.97 0.64 -0.49] [0. 0.04 0.96] [-3.22 2.37 -0.55]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.71 2.28 -1.26]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.2 -1.46]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.15 -1.5 ]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.87 2.12 -1.48]
C [-0.19 0.54 -0.86] [0.05 0.01 0.94] [-0.19 2.11 -1.4 ]
C [ 0.51 0.57 -0.8 ] [0.86 0. 0.14] [ 0.58 2.13 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.39 0.58]
A [0.51 0.72 0.65] [0.56 0.44 0. ] [0.65 2.59 1.09]
A [0.41 0.72 0.76] [0.34 0.66 0. ] [0.49 2.76 1.5 ]
A [0.32 0.72 0.82] [0.2 0.79 0. ] [0.37 2.89 1.88]
B [-0.54 0.7 0.89] [0. 1. 0.] [-0.65 2.82 1.99]
B [-0.89 0.66 0.93] [0. 1. 0.] [-1.58 2.72 1.92]
B [-0.96 0.64 0.92] [0. 1. 0.] [-2.5 2.6 1.74]
B [-0.97 0.64 0.89] [0. 1. 0.] [-3.41 2.49 1.51]
B [-0.97 0.63 0.83] [0. 1. 0.] [-4.31 2.4 1.24]
B [-0.97 0.63 0.69] [0. 0.99 0.01] [-5.2 2.32 0.87]
B [-0.97 0.63 0.32] [0. 0.93 0.07] [-6.07 2.25 0.34]
B [-0.97 0.63 -0.43] [0. 0.06 0.94] [-6.93 2.2 -0.48]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-6.39 2.15 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-5.75 2.1 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-5.12 2.07 -1.51]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-4.49 2.05 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.86 2.03 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.24 2.02 -1.52]
C [-0.97 0.53 -0.89] [0. 0. 1.] [-2.62 2.01 -1.52]
C [-0.95 0.53 -0.89] [0. 0. 1.] [-2.01 2. -1.52]
C [-0.87 0.53 -0.88] [0. 0. 1.] [-1.4 2. -1.51]
C [-0.63 0.53 -0.88] [0. 0. 0.99] [-0.76 2. -1.48]
C [-0.07 0.54 -0.85] [0.1 0.01 0.89] [-0.07 2.02 -1.38]
C [ 0.6 0.57 -0.78] [0.92 0. 0.08] [ 0.72 2.07 -1.19]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.34 0.61]
A [0.52 0.72 0.65] [0.57 0.43 0. ] [0.66 2.56 1.1 ]
B [-0.41 0.71 0.72] [0.01 0.99 0. ] [-0.47 2.54 1.07]
B [-0.85 0.67 0.65] [0. 0.99 0.01] [-1.39 2.49 0.82]
B [-0.95 0.64 0.29] [0. 0.91 0.08] [-2.31 2.41 0.3 ]
B [-0.97 0.64 -0.46] [0. 0.05 0.95] [-3.21 2.35 -0.52]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.71 2.26 -1.25]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.19 -1.45]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.14 -1.49]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.87 2.11 -1.48]
C [-0.18 0.54 -0.86] [0.05 0.01 0.94] [-0.19 2.1 -1.4 ]
C [ 0.51 0.57 -0.8 ] [0.86 0. 0.14] [ 0.58 2.12 -1.23]
epoch: 0
loss: 0.0124
-----
```

```

-----
symbol= AAABBBBBBCCCCCCCCABBBCCABBBCCAABBBBBCCCCCAABBBBBBCCCCCCCCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
A [0.48 0.49 0.72] [0.51 0.49 0. ] [0.63 0.84 1.25]
A [0.46 0.6 0.78] [0.4 0.6 0. ] [0.59 1.2 1.64]
B [-0.43 0.63 0.85] [0.01 0.99 0. ] [-0.5 1.38 1.68]
B [-0.86 0.6 0.88] [0. 1. 0.] [-1.42 1.47 1.56]
B [-0.95 0.59 0.84] [0. 1. 0.] [-2.34 1.52 1.3 ]
B [-0.97 0.59 0.72] [0. 0.99 0. ] [-3.25 1.56 0.94]
B [-0.97 0.6 0.39] [0. 0.95 0.05] [-4.15 1.6 0.42]
B [-0.97 0.6 -0.34] [0. 0.1 0.9] [-5.03 1.64 -0.37]
C [-0.98 0.51 -0.81] [0. 0. 1.] [-4.51 1.71 -1.18]
C [-0.98 0.51 -0.87] [0. 0. 1.] [-3.89 1.77 -1.44]
C [-0.98 0.52 -0.88] [0. 0. 1.] [-3.27 1.81 -1.5 ]
C [-0.97 0.52 -0.88] [0. 0. 1.] [-2.65 1.85 -1.52]
C [-0.95 0.52 -0.88] [0. 0. 1.] [-2.04 1.88 -1.52]
C [-0.87 0.52 -0.88] [0. 0. 1.] [-1.42 1.91 -1.51]
C [-0.65 0.53 -0.88] [0. 0. 0.99] [-0.79 1.93 -1.48]
C [-0.1 0.54 -0.85] [0.09 0.01 0.91] [-0.1 1.97 -1.39]
C [ 0.58 0.57 -0.79] [0.9 0. 0.09] [ 0.68 2.02 -1.2 ]
A [0.62 0.72 0.38] [0.85 0.15 0. ] [0.87 2.31 0.6 ]
B [-0.32 0.71 0.23] [0.07 0.88 0.05] [-0.35 2.34 0.26]
B [-0.82 0.67 -0.41] [0. 0.08 0.91] [-1.27 2.33 -0.46]
C [-0.62 0.54 -0.82] [0. 0.01 0.99] [-0.74 2.26 -1.2 ]
C [-0.05 0.55 -0.84] [0.12 0.01 0.87] [-0.05 2.22 -1.33]
C [ 0.61 0.58 -0.78] [0.92 0. 0.07] [ 0.74 2.22 -1.18]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.48 0.61]
B [-0.31 0.72 0.24] [0.07 0.89 0.04] [-0.35 2.48 0.28]
B [-0.82 0.67 -0.39] [0. 0.1 0.9] [-1.26 2.46 -0.44]
C [-0.62 0.54 -0.81] [0. 0.01 0.99] [-0.74 2.36 -1.19]
C [-0.05 0.55 -0.84] [0.12 0.01 0.87] [-0.05 2.3 -1.33]
C [ 0.61 0.58 -0.78] [0.92 0. 0.07] [ 0.74 2.27 -1.18]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.52 0.61]
A [0.52 0.73 0.65] [0.56 0.44 0. ] [0.66 2.71 1.1 ]
B [-0.41 0.71 0.72] [0.01 0.99 0. ] [-0.47 2.68 1.07]
B [-0.85 0.67 0.65] [0. 0.99 0.01] [-1.39 2.61 0.83]
B [-0.95 0.65 0.3 ] [0. 0.92 0.08] [-2.31 2.51 0.32]
B [-0.97 0.64 -0.45] [0. 0.05 0.95] [-3.22 2.43 -0.5 ]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.72 2.33 -1.24]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.24 -1.45]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.18 -1.49]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.88 2.14 -1.48]
C [-0.19 0.54 -0.86] [0.05 0.01 0.94] [-0.2 2.12 -1.4 ]
C [ 0.5 0.57 -0.8 ] [0.85 0. 0.14] [ 0.57 2.14 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.4 0.58]
A [0.51 0.72 0.65] [0.56 0.44 0. ] [0.65 2.6 1.09]
A [0.41 0.72 0.76] [0.33 0.67 0. ] [0.49 2.77 1.5 ]
B [-0.48 0.71 0.84] [0. 1. 0.] [-0.57 2.72 1.58]
B [-0.87 0.66 0.87] [0. 1. 0.] [-1.49 2.63 1.48]
B [-0.96 0.64 0.82] [0. 1. 0.] [-2.42 2.53 1.23]
B [-0.97 0.64 0.69] [0. 0.99 0.01] [-3.33 2.43 0.87]
B [-0.97 0.64 0.32] [0. 0.93 0.07] [-4.23 2.35 0.34]
B [-0.97 0.64 -0.43] [0. 0.06 0.94] [-5.11 2.29 -0.47]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-4.59 2.21 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-3.97 2.16 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-3.35 2.11 -1.5 ]
C [-0.97 0.53 -0.89] [0. 0. 1.] [-2.74 2.08 -1.52]
C [-0.96 0.53 -0.89] [0. 0. 1.] [-2.13 2.06 -1.52]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.51 2.04 -1.51]
C [-0.7 0.53 -0.88] [0. 0. 1.] [-0.89 2.03 -1.48]
C [-0.2 0.54 -0.86] [0.05 0.01 0.95] [-0.21 2.04 -1.41]
C [ 0.5 0.57 -0.8 ] [0.84 0. 0.15] [ 0.56 2.08 -1.23]
epoch: 1
loss: 0.0091
-----

```



```

-----
symbol= AAABBBBBBCCCCCCCCAAABBBBBBCCCCCCCCABBCCCABBCCCAABBBBBCCCCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
A [0.48 0.49 0.72] [0.51 0.49 0. ] [0.63 0.84 1.25]
A [0.46 0.6 0.78] [0.4 0.6 0. ] [0.59 1.2 1.64]
B [-0.43 0.63 0.85] [0.01 0.99 0. ] [-0.5 1.38 1.68]
B [-0.86 0.6 0.88] [0. 1. 0.] [-1.42 1.47 1.56]
B [-0.95 0.59 0.84] [0. 1. 0.] [-2.34 1.52 1.3 ]
B [-0.97 0.59 0.72] [0. 0.99 0. ] [-3.25 1.56 0.94]
B [-0.97 0.6 0.39] [0. 0.95 0.05] [-4.15 1.6 0.42]
B [-0.97 0.6 -0.34] [0. 0.1 0.9] [-5.03 1.64 -0.37]
C [-0.98 0.51 -0.81] [0. 0. 1.] [-4.51 1.71 -1.18]
C [-0.98 0.51 -0.87] [0. 0. 1.] [-3.89 1.77 -1.44]
C [-0.98 0.52 -0.88] [0. 0. 1.] [-3.27 1.81 -1.5 ]
C [-0.97 0.52 -0.88] [0. 0. 1.] [-2.65 1.85 -1.52]
C [-0.95 0.52 -0.88] [0. 0. 1.] [-2.04 1.88 -1.52]
C [-0.87 0.52 -0.88] [0. 0. 1.] [-1.42 1.91 -1.51]
C [-0.65 0.53 -0.88] [0. 0. 0.99] [-0.79 1.93 -1.48]
C [-0.1 0.54 -0.85] [0.09 0.01 0.91] [-0.1 1.97 -1.39]
C [ 0.58 0.57 -0.79] [0.9 0. 0.09] [ 0.68 2.02 -1.2 ]
A [0.62 0.72 0.38] [0.85 0.15 0. ] [0.87 2.31 0.6 ]
A [0.52 0.72 0.65] [0.56 0.43 0. ] [0.66 2.52 1.1 ]
A [0.41 0.72 0.76] [0.34 0.66 0. ] [0.5 2.69 1.5 ]
B [-0.48 0.71 0.84] [0. 1. 0.] [-0.56 2.66 1.58]
B [-0.87 0.66 0.87] [0. 1. 0.] [-1.49 2.58 1.48]
B [-0.96 0.64 0.82] [0. 1. 0.] [-2.41 2.48 1.23]
B [-0.97 0.64 0.69] [0. 0.99 0.01] [-3.32 2.39 0.87]
B [-0.97 0.63 0.32] [0. 0.93 0.07] [-4.22 2.32 0.34]
B [-0.97 0.64 -0.43] [0. 0.06 0.94] [-5.1 2.26 -0.47]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-4.59 2.19 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-3.97 2.14 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-3.35 2.1 -1.5 ]
C [-0.97 0.53 -0.89] [0. 0. 1.] [-2.73 2.07 -1.52]
C [-0.95 0.53 -0.89] [0. 0. 1.] [-2.12 2.05 -1.52]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.51 2.03 -1.51]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.88 2.03 -1.48]
C [-0.2 0.54 -0.86] [0.05 0.01 0.95] [-0.2 2.04 -1.41]
C [ 0.5 0.57 -0.8 ] [0.85 0. 0.15] [ 0.57 2.07 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.34 0.58]
B [-0.33 0.71 0.21] [0.07 0.88 0.05] [-0.37 2.37 0.24]
B [-0.83 0.67 -0.44] [0. 0.07 0.93] [-1.28 2.36 -0.5 ]
C [-0.62 0.54 -0.82] [0. 0.01 0.99] [-0.75 2.29 -1.22]
C [-0.06 0.55 -0.84] [0.11 0.01 0.88] [-0.06 2.24 -1.33]
C [ 0.6 0.58 -0.78] [0.92 0. 0.08] [ 0.72 2.23 -1.19]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.49 0.61]
B [-0.31 0.72 0.24] [0.07 0.89 0.04] [-0.35 2.49 0.28]
B [-0.82 0.67 -0.39] [0. 0.09 0.9 ] [-1.26 2.47 -0.44]
C [-0.62 0.54 -0.81] [0. 0.01 0.99] [-0.74 2.37 -1.19]
C [-0.05 0.55 -0.84] [0.12 0.01 0.87] [-0.05 2.3 -1.33]
C [ 0.61 0.58 -0.78] [0.92 0. 0.07] [ 0.73 2.28 -1.18]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.53 0.61]
A [0.52 0.73 0.65] [0.56 0.44 0. ] [0.66 2.72 1.1 ]
B [-0.41 0.71 0.72] [0.01 0.99 0. ] [-0.47 2.68 1.07]
B [-0.85 0.67 0.65] [0. 0.99 0.01] [-1.39 2.61 0.83]
B [-0.95 0.65 0.3 ] [0. 0.92 0.08] [-2.31 2.52 0.32]
B [-0.97 0.64 -0.45] [0. 0.05 0.95] [-3.22 2.43 -0.5 ]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.72 2.33 -1.24]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.24 -1.45]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.18 -1.49]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.88 2.14 -1.48]
C [-0.19 0.54 -0.86] [0.05 0.01 0.94] [-0.2 2.12 -1.4 ]
C [ 0.5 0.57 -0.8 ] [0.85 0. 0.14] [ 0.57 2.14 -1.23]
epoch: 2
loss: 0.0090
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symbol= AAAABBBBBBBBBCCCCCCCCCCCCAABBBBCCCCCABBCCCAABBBBCCCCCAABBBBCCCCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
A [0.48 0.49 0.72] [0.51 0.49 0. ] [0.63 0.84 1.25]
A [0.46 0.6 0.78] [0.4 0.6 0. ] [0.59 1.2 1.64]
A [0.4 0.66 0.82] [0.29 0.71 0. ] [0.49 1.52 1.96]
B [-0.48 0.66 0.89] [0. 1. 0.] [-0.57 1.65 2.04]
B [-0.87 0.63 0.93] [0. 1. 0.] [-1.49 1.7 1.97]
B [-0.96 0.61 0.92] [0. 1. 0.] [-2.41 1.72 1.77]
B [-0.97 0.61 0.89] [0. 1. 0.] [-3.32 1.73 1.53]
B [-0.97 0.61 0.82] [0. 1. 0.] [-4.22 1.74 1.23]
B [-0.97 0.61 0.68] [0. 0.99 0.01] [-5.11 1.75 0.85]
B [-0.97 0.61 0.28] [0. 0.9 0.09] [-5.98 1.76 0.3 ]
B [-0.97 0.61 -0.47] [0. 0.04 0.96] [-6.83 1.78 -0.54]
C [-0.98 0.52 -0.83] [0. 0. 1.] [-6.28 1.82 -1.26]
C [-0.98 0.52 -0.87] [0. 0. 1.] [-5.64 1.86 -1.46]
C [-0.98 0.52 -0.88] [0. 0. 1.] [-5. 1.88 -1.51]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-4.37 1.9 -1.52]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-3.74 1.92 -1.52]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-3.12 1.93 -1.52]
C [-0.97 0.52 -0.89] [0. 0. 1.] [-2.51 1.94 -1.52]
C [-0.94 0.52 -0.88] [0. 0. 1.] [-1.89 1.95 -1.52]
C [-0.84 0.53 -0.88] [0. 0. 1.] [-1.28 1.96 -1.5 ]
C [-0.55 0.53 -0.87] [0. 0. 0.99] [-0.64 1.97 -1.46]
C [ 0.07 0.55 -0.84] [0.23 0.01 0.76] [ 0.08 2.01 -1.35]
C [ 0.68 0.58 -0.77] [0.95 0. 0.05] [ 0.86 2.06 -1.15]
A [0.64 0.73 0.38] [0.85 0.14 0. ] [0.91 2.34 0.63]
A [0.53 0.73 0.65] [0.57 0.43 0. ] [0.67 2.55 1.11]
B [-0.41 0.71 0.72] [0.01 0.99 0. ] [-0.46 2.54 1.08]
B [-0.85 0.67 0.66] [0. 0.99 0.01] [-1.38 2.49 0.84]
B [-0.95 0.64 0.31] [0. 0.92 0.07] [-2.3 2.41 0.33]
B [-0.97 0.64 -0.44] [0. 0.05 0.95] [-3.21 2.35 -0.49]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.71 2.26 -1.24]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.19 -1.45]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.14 -1.49]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.87 2.11 -1.48]
C [-0.18 0.54 -0.86] [0.05 0.01 0.94] [-0.19 2.1 -1.4 ]
C [ 0.51 0.57 -0.8 ] [0.86 0. 0.14] [ 0.58 2.12 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.39 0.58]
B [-0.33 0.71 0.21] [0.07 0.88 0.05] [-0.37 2.41 0.24]
B [-0.83 0.67 -0.43] [0. 0.07 0.93] [-1.28 2.39 -0.49]
C [-0.62 0.54 -0.82] [0. 0.01 0.99] [-0.75 2.31 -1.22]
C [-0.06 0.55 -0.84] [0.11 0.01 0.88] [-0.06 2.26 -1.33]
C [ 0.6 0.58 -0.78] [0.92 0. 0.08] [ 0.72 2.24 -1.19]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.5 0.61]
A [0.52 0.73 0.65] [0.56 0.44 0. ] [0.66 2.69 1.1 ]
B [-0.41 0.71 0.72] [0.01 0.99 0. ] [-0.47 2.66 1.07]
B [-0.85 0.67 0.65] [0. 0.99 0.01] [-1.39 2.59 0.83]
B [-0.95 0.64 0.3 ] [0. 0.92 0.08] [-2.31 2.5 0.31]
B [-0.97 0.64 -0.45] [0. 0.05 0.95] [-3.22 2.42 -0.5 ]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.71 2.32 -1.24]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.23 -1.45]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.17 -1.49]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.88 2.14 -1.48]
C [-0.19 0.54 -0.86] [0.05 0.01 0.94] [-0.2 2.12 -1.4 ]
C [ 0.5 0.57 -0.8 ] [0.85 0. 0.14] [ 0.57 2.14 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.4 0.58]
A [0.51 0.72 0.65] [0.56 0.44 0. ] [0.65 2.6 1.09]
B [-0.42 0.71 0.71] [0.01 0.99 0. ] [-0.47 2.58 1.06]
B [-0.85 0.67 0.64] [0. 0.99 0.01] [-1.4 2.52 0.81]
B [-0.95 0.64 0.27] [0. 0.9 0.1] [-2.32 2.44 0.28]
B [-0.97 0.64 -0.49] [0. 0.04 0.96] [-3.22 2.37 -0.55]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.71 2.28 -1.26]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.21 -1.46]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.15 -1.5 ]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.87 2.12 -1.48]
C [-0.19 0.54 -0.86] [0.05 0.01 0.94] [-0.19 2.11 -1.4 ]
C [ 0.51 0.57 -0.8 ] [0.86 0. 0.14] [ 0.58 2.13 -1.23]
epoch: 3
loss: 0.0095
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-----
symbol= AAABBBBBBBBBCCCCCCCCABBBCCCAABBBBCCCCCAABBBBCCCCCCCCCAAAABBBBCCCCCCCCCCCCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
A [0.48 0.49 0.72] [0.51 0.49 0. ] [0.63 0.84 1.25]
A [0.46 0.6 0.78] [0.4 0.6 0. ] [0.59 1.2 1.64]
B [-0.43 0.63 0.85] [0.01 0.99 0. ] [-0.5 1.38 1.68]
B [-0.86 0.6 0.88] [0. 1. 0. ] [-1.42 1.47 1.56]
B [-0.95 0.59 0.84] [0. 1. 0. ] [-2.34 1.52 1.3 ]
B [-0.97 0.59 0.72] [0. 0.99 0. ] [-3.25 1.56 0.94]
B [-0.97 0.6 0.39] [0. 0.95 0.05] [-4.15 1.6 0.42]
B [-0.97 0.6 -0.34] [0. 0.1 0.9] [-5.03 1.64 -0.37]
C [-0.98 0.51 -0.81] [0. 0. 1.] [-4.51 1.71 -1.18]
C [-0.98 0.51 -0.87] [0. 0. 1.] [-3.89 1.77 -1.44]
C [-0.98 0.52 -0.88] [0. 0. 1.] [-3.27 1.81 -1.5 ]
C [-0.97 0.52 -0.88] [0. 0. 1.] [-2.65 1.85 -1.52]
C [-0.95 0.52 -0.88] [0. 0. 1.] [-2.04 1.88 -1.52]
C [-0.87 0.52 -0.88] [0. 0. 1.] [-1.42 1.91 -1.51]
C [-0.65 0.53 -0.88] [0. 0. 0.99] [-0.79 1.93 -1.48]
C [-0.1 0.54 -0.85] [0.09 0.01 0.91] [-0.1 1.97 -1.39]
C [ 0.58 0.57 -0.79] [0.9 0. 0.09] [ 0.68 2.02 -1.2 ]
A [0.62 0.72 0.38] [0.85 0.15 0. ] [0.87 2.31 0.6 ]
B [-0.32 0.71 0.23] [0.07 0.88 0.05] [-0.35 2.34 0.26]
B [-0.82 0.67 -0.41] [0. 0.08 0.91] [-1.27 2.33 -0.46]
C [-0.62 0.54 -0.82] [0. 0.01 0.99] [-0.74 2.26 -1.2 ]
C [-0.05 0.55 -0.84] [0.12 0.01 0.87] [-0.05 2.22 -1.33]
C [ 0.61 0.58 -0.78] [0.92 0. 0.07] [ 0.74 2.22 -1.18]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.48 0.61]
A [0.52 0.73 0.65] [0.56 0.44 0. ] [0.66 2.67 1.1 ]
B [-0.41 0.71 0.72] [0.01 0.99 0. ] [-0.47 2.64 1.07]
B [-0.85 0.67 0.65] [0. 0.99 0.01] [-1.39 2.58 0.83]
B [-0.95 0.64 0.3 ] [0. 0.92 0.08] [-2.31 2.49 0.32]
B [-0.97 0.64 -0.45] [0. 0.05 0.95] [-3.22 2.41 -0.5 ]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.71 2.31 -1.24]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.23 -1.45]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.17 -1.49]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.88 2.13 -1.48]
C [-0.19 0.54 -0.86] [0.05 0.01 0.94] [-0.2 2.12 -1.4 ]
C [ 0.51 0.57 -0.8 ] [0.85 0. 0.14] [ 0.58 2.14 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.4 0.58]
A [0.51 0.72 0.65] [0.56 0.44 0. ] [0.65 2.6 1.09]
A [0.41 0.72 0.76] [0.33 0.66 0. ] [0.49 2.76 1.5 ]
B [-0.48 0.71 0.84] [0. 1. 0. ] [-0.57 2.72 1.58]
B [-0.87 0.66 0.87] [0. 1. 0. ] [-1.49 2.63 1.48]
B [-0.96 0.64 0.82] [0. 1. 0. ] [-2.42 2.53 1.23]
B [-0.97 0.64 0.69] [0. 0.99 0.01] [-3.33 2.43 0.87]
B [-0.97 0.64 0.32] [0. 0.93 0.07] [-4.23 2.35 0.34]
B [-0.97 0.64 -0.43] [0. 0.06 0.94] [-5.11 2.29 -0.47]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-4.59 2.21 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-3.97 2.16 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-3.35 2.11 -1.5 ]
C [-0.97 0.53 -0.89] [0. 0. 1.] [-2.74 2.08 -1.52]
C [-0.96 0.53 -0.89] [0. 0. 1.] [-2.13 2.06 -1.52]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.51 2.04 -1.51]
C [-0.7 0.53 -0.88] [0. 0. 1.] [-0.89 2.03 -1.48]
C [-0.2 0.54 -0.86] [0.05 0.01 0.95] [-0.21 2.04 -1.41]
C [ 0.5 0.57 -0.8 ] [0.84 0. 0.15] [ 0.56 2.08 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.34 0.58]
A [0.51 0.72 0.65] [0.56 0.44 0. ] [0.65 2.55 1.09]
A [0.41 0.72 0.76] [0.34 0.66 0. ] [0.49 2.72 1.5 ]
A [0.32 0.72 0.82] [0.21 0.79 0. ] [0.37 2.86 1.88]
B [-0.54 0.7 0.89] [0. 1. 0. ] [-0.65 2.79 1.98]
B [-0.89 0.66 0.93] [0. 1. 0. ] [-1.58 2.69 1.92]
B [-0.96 0.64 0.92] [0. 1. 0. ] [-2.5 2.58 1.74]
B [-0.97 0.64 0.89] [0. 1. 0. ] [-3.41 2.47 1.51]
B [-0.97 0.63 0.82] [0. 1. 0. ] [-4.31 2.38 1.23]
B [-0.97 0.63 0.68] [0. 0.99 0.01] [-5.2 2.3 0.87]
B [-0.97 0.63 0.31] [0. 0.92 0.07] [-6.07 2.24 0.33]
B [-0.97 0.63 -0.43] [0. 0.05 0.94] [-6.93 2.19 -0.48]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-6.39 2.14 -1.24]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-5.75 2.1 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-5.12 2.07 -1.51]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-4.48 2.05 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.86 2.03 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.24 2.02 -1.52]
C [-0.97 0.53 -0.89] [0. 0. 1.] [-2.62 2.01 -1.52]
C [-0.95 0.53 -0.89] [0. 0. 1.] [-2.01 2. -1.52]
C [-0.87 0.53 -0.88] [0. 0. 1.] [-1.39 2. -1.51]
C [-0.63 0.53 -0.88] [0. 0. 0.99] [-0.76 2. -1.48]
C [-0.06 0.54 -0.85] [0.11 0.01 0.89] [-0.07 2.02 -1.38]
C [ 0.6 0.57 -0.78] [0.92 0. 0.08] [ 0.72 2.07 -1.19]
epoch: 4
loss: 0.0123
-----
```

```
-----
symbol= AAAABBBBBBBBCCCCCCCCCCCCAAAABBBBBBBBCCCCCCCCCCCCAAABBBBBBCCCCCCCCABBBCCABBBCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
A [0.48 0.49 0.72] [0.51 0.49 0. ] [0.63 0.84 1.25]
A [0.46 0.6 0.78] [0.4 0.6 0. ] [0.59 1.2 1.64]
A [0.4 0.66 0.82] [0.29 0.71 0. ] [0.49 1.52 1.96]
B [-0.48 0.66 0.89] [0. 1. 0.] [-0.57 1.65 2.04]
B [-0.87 0.63 0.93] [0. 1. 0.] [-1.49 1.7 1.97]
B [-0.96 0.61 0.92] [0. 1. 0.] [-2.41 1.72 1.77]
B [-0.97 0.61 0.89] [0. 1. 0.] [-3.32 1.73 1.53]
B [-0.97 0.61 0.82] [0. 1. 0.] [-4.22 1.74 1.23]
B [-0.97 0.61 0.68] [0. 0.99 0.01] [-5.11 1.75 0.85]
B [-0.97 0.61 0.28] [0. 0.9 0.09] [-5.98 1.76 0.3 ]
B [-0.97 0.61 -0.47] [0. 0.04 0.96] [-6.83 1.78 -0.54]
C [-0.98 0.52 -0.83] [0. 0. 1.] [-6.28 1.82 -1.26]
C [-0.98 0.52 -0.87] [0. 0. 1.] [-5.64 1.86 -1.46]
C [-0.98 0.52 -0.88] [0. 0. 1.] [-5. 1.88 -1.51]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-4.37 1.9 -1.52]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-3.74 1.92 -1.52]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-3.12 1.93 -1.52]
C [-0.97 0.52 -0.89] [0. 0. 1.] [-2.51 1.94 -1.52]
C [-0.94 0.52 -0.88] [0. 0. 1.] [-1.89 1.95 -1.52]
C [-0.84 0.53 -0.88] [0. 0. 1.] [-1.28 1.96 -1.5 ]
C [-0.55 0.53 -0.87] [0. 0. 0.99] [-0.64 1.97 -1.46]
C [ 0.07 0.55 -0.84] [0.23 0.01 0.76] [ 0.08 2.01 -1.35]
C [ 0.68 0.58 -0.77] [0.95 0. 0.05] [ 0.86 2.06 -1.15]
A [0.64 0.73 0.38] [0.85 0.14 0. ] [0.91 2.34 0.63]
A [0.53 0.73 0.65] [0.57 0.43 0. ] [0.67 2.55 1.11]
A [0.42 0.72 0.76] [0.34 0.66 0. ] [0.5 2.72 1.51]
A [0.32 0.72 0.82] [0.21 0.79 0. ] [0.37 2.86 1.88]
B [-0.54 0.7 0.89] [0. 1. 0.] [-0.64 2.8 1.98]
B [-0.89 0.66 0.93] [0. 1. 0.] [-1.57 2.7 1.92]
B [-0.96 0.64 0.92] [0. 1. 0.] [-2.49 2.58 1.74]
B [-0.97 0.64 0.89] [0. 1. 0.] [-3.41 2.47 1.51]
B [-0.97 0.63 0.82] [0. 1. 0.] [-4.31 2.38 1.23]
B [-0.97 0.63 0.69] [0. 0.99 0.01] [-5.19 2.3 0.87]
B [-0.97 0.63 0.32] [0. 0.93 0.07] [-6.07 2.24 0.34]
B [-0.97 0.63 -0.43] [0. 0.06 0.94] [-6.92 2.19 -0.48]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-6.38 2.14 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-5.75 2.1 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-5.11 2.07 -1.51]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-4.48 2.05 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.85 2.03 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.23 2.02 -1.52]
C [-0.97 0.53 -0.89] [0. 0. 1.] [-2.62 2.01 -1.52]
C [-0.95 0.53 -0.89] [0. 0. 1.] [-2.01 2. -1.52]
C [-0.87 0.53 -0.88] [0. 0. 1.] [-1.39 2. -1.51]
C [-0.63 0.53 -0.88] [0. 0. 0.99] [-0.76 2. -1.48]
C [-0.06 0.54 -0.85] [0.11 0.01 0.88] [-0.06 2.02 -1.38]
C [ 0.6 0.58 -0.78] [0.92 0. 0.08] [ 0.72 2.07 -1.19]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.34 0.61]
A [0.52 0.72 0.65] [0.57 0.43 0. ] [0.66 2.56 1.1 ]
A [0.42 0.72 0.76] [0.34 0.66 0. ] [0.5 2.73 1.51]
B [-0.48 0.71 0.84] [0. 1. 0.] [-0.56 2.68 1.58]
B [-0.87 0.66 0.87] [0. 1. 0.] [-1.49 2.6 1.48]
B [-0.96 0.64 0.82] [0. 1. 0.] [-2.41 2.5 1.23]
B [-0.97 0.64 0.69] [0. 0.99 0.01] [-3.32 2.41 0.88]
B [-0.97 0.63 0.33] [0. 0.93 0.07] [-4.22 2.33 0.35]
B [-0.97 0.64 -0.42] [0. 0.06 0.94] [-5.1 2.27 -0.46]
C [-0.98 0.53 -0.82] [0. 0. 1.] [-4.59 2.2 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-3.97 2.15 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-3.35 2.1 -1.5 ]
C [-0.97 0.53 -0.88] [0. 0. 1.] [-2.74 2.07 -1.52]
C [-0.95 0.53 -0.89] [0. 0. 1.] [-2.12 2.05 -1.52]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.51 2.03 -1.51]
C [-0.7 0.53 -0.88] [0. 0. 1.] [-0.88 2.03 -1.48]
C [-0.2 0.54 -0.86] [0.05 0.01 0.95] [-0.2 2.04 -1.41]
C [ 0.5 0.57 -0.8 ] [0.85 0. 0.15] [ 0.57 2.07 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.83 2.34 0.58]
B [-0.33 0.71 0.21] [0.07 0.88 0.05] [-0.37 2.37 0.24]
B [-0.83 0.67 -0.44] [0. 0.07 0.93] [-1.28 2.36 -0.5 ]
C [-0.63 0.54 -0.82] [0. 0.01 0.99] [-0.75 2.29 -1.22]
C [-0.06 0.55 -0.84] [0.11 0.01 0.88] [-0.06 2.24 -1.33]
C [ 0.6 0.58 -0.78] [0.92 0. 0.08] [ 0.72 2.23 -1.19]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.49 0.61]
B [-0.31 0.72 0.24] [0.07 0.89 0.04] [-0.35 2.49 0.28]
B [-0.82 0.67 -0.39] [0. 0.09 0.9 ] [-1.26 2.47 -0.44]
C [-0.62 0.54 -0.81] [0. 0.01 0.99] [-0.74 2.37 -1.19]
C [-0.05 0.55 -0.84] [0.12 0.01 0.87] [-0.05 2.3 -1.33]
C [ 0.61 0.58 -0.78] [0.92 0. 0.07] [ 0.73 2.28 -1.18]
epoch: 5
loss: 0.0141
-----
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-----
symbol= ABBBBBCCCCCABBCCCAAABBBBBBCCCCCCCCCABBCCCAAABBBBBBCCCCCCCCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
A [0.48 0.49 0.72] [0.51 0.49 0. ] [0.63 0.84 1.25]
B [-0.4 0.56 0.75] [0.01 0.99 0. ] [-0.46 1.07 1.17]
B [-0.84 0.56 0.68] [0. 0.99 0.01] [-1.37 1.21 0.89]
B [-0.95 0.56 0.33] [0. 0.93 0.07] [-2.29 1.3 0.36]
B [-0.97 0.57 -0.41] [0. 0.06 0.94] [-3.19 1.38 -0.46]
C [-0.97 0.49 -0.82] [0. 0. 1.] [-2.67 1.51 -1.22]
C [-0.95 0.5 -0.87] [0. 0. 1.] [-2.05 1.62 -1.45]
C [-0.88 0.51 -0.88] [0. 0. 1.] [-1.44 1.7 -1.49]
C [-0.65 0.52 -0.88] [0. 0. 0.99] [-0.8 1.78 -1.48]
C [-0.11 0.54 -0.85] [0.08 0.01 0.91] [-0.11 1.85 -1.39]
C [ 0.57 0.57 -0.79] [0.9 0. 0.1] [ 0.67 1.93 -1.2 ]
A [0.62 0.72 0.38] [0.85 0.15 0. ] [0.87 2.23 0.6 ]
B [-0.32 0.71 0.22] [0.07 0.88 0.05] [-0.35 2.27 0.26]
B [-0.82 0.67 -0.42] [0. 0.08 0.92] [-1.27 2.28 -0.47]
C [-0.62 0.54 -0.82] [0. 0.01 0.99] [-0.74 2.22 -1.21]
C [-0.04 0.55 -0.84] [0.12 0.01 0.87] [-0.05 2.19 -1.33]
C [ 0.61 0.58 -0.78] [0.92 0. 0.07] [ 0.74 2.19 -1.18]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.45 0.61]
A [0.52 0.73 0.65] [0.57 0.43 0. ] [0.66 2.65 1.1 ]
A [0.41 0.72 0.76] [0.34 0.66 0. ] [0.5 2.81 1.51]
B [-0.48 0.71 0.84] [0. 1. 0.] [-0.56 2.76 1.58]
B [-0.87 0.66 0.87] [0. 1. 0.] [-1.49 2.66 1.48]
B [-0.96 0.64 0.82] [0. 1. 0.] [-2.41 2.56 1.24]
B [-0.97 0.64 0.69] [0. 0.99 0.01] [-3.32 2.46 0.88]
B [-0.97 0.64 0.34] [0. 0.93 0.06] [-4.22 2.37 0.36]
B [-0.97 0.64 -0.41] [0. 0.07 0.93] [-5.1 2.31 -0.45]
C [-0.98 0.53 -0.82] [0. 0. 1.] [-4.59 2.23 -1.22]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-3.98 2.17 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-3.36 2.12 -1.5 ]
C [-0.97 0.53 -0.88] [0. 0. 1.] [-2.74 2.09 -1.52]
C [-0.96 0.53 -0.89] [0. 0. 1.] [-2.13 2.06 -1.52]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.52 2.04 -1.51]
C [-0.7 0.53 -0.88] [0. 0. 1.] [-0.89 2.03 -1.49]
C [-0.2 0.54 -0.86] [0.04 0.01 0.95] [-0.21 2.04 -1.41]
C [ 0.49 0.57 -0.8 ] [0.84 0. 0.15] [ 0.56 2.08 -1.23]
A [0.61 0.72 0.37] [0.84 0.16 0. ] [0.83 2.35 0.58]
B [-0.33 0.71 0.2 ] [0.07 0.88 0.05] [-0.37 2.37 0.23]
B [-0.83 0.67 -0.44] [0. 0.07 0.93] [-1.28 2.36 -0.51]
C [-0.63 0.54 -0.82] [0. 0.01 0.99] [-0.75 2.29 -1.22]
C [-0.06 0.55 -0.84] [0.11 0.01 0.88] [-0.06 2.24 -1.33]
C [ 0.6 0.58 -0.78] [0.92 0. 0.08] [ 0.72 2.23 -1.19]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.49 0.61]
A [0.52 0.73 0.65] [0.56 0.44 0. ] [0.66 2.68 1.1 ]
A [0.41 0.72 0.76] [0.34 0.66 0. ] [0.5 2.84 1.51]
B [-0.48 0.71 0.84] [0. 1. 0.] [-0.56 2.78 1.58]
B [-0.87 0.66 0.87] [0. 1. 0.] [-1.49 2.68 1.48]
B [-0.96 0.64 0.82] [0. 1. 0.] [-2.41 2.57 1.24]
B [-0.97 0.64 0.69] [0. 0.99 0.01] [-3.33 2.47 0.88]
B [-0.97 0.64 0.34] [0. 0.94 0.06] [-4.22 2.38 0.36]
B [-0.97 0.64 -0.41] [0. 0.07 0.93] [-5.1 2.32 -0.45]
C [-0.98 0.53 -0.82] [0. 0. 1.] [-4.59 2.24 -1.22]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-3.98 2.17 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-3.36 2.12 -1.5 ]
C [-0.97 0.53 -0.88] [0. 0. 1.] [-2.74 2.09 -1.52]
C [-0.96 0.53 -0.89] [0. 0. 1.] [-2.13 2.06 -1.52]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.52 2.04 -1.51]
C [-0.7 0.53 -0.88] [0. 0. 1.] [-0.89 2.03 -1.49]
C [-0.21 0.54 -0.86] [0.04 0.01 0.95] [-0.21 2.04 -1.41]
C [ 0.49 0.57 -0.8 ] [0.84 0. 0.15] [ 0.56 2.08 -1.24]
epoch: 6
loss: 0.0096
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-----
symbol= AAAABBBBCCCCCCCCCAAAABBBBCCCCCCCCCAABBBBCCCCCCCCCAAAABBBBCCCCCCCCCAAAABBBBCCCCCCCCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
A [0.48 0.49 0.72] [0.51 0.49 0. ] [0.63 0.84 1.25]
A [0.46 0.6 0.78] [0.4 0.6 0. ] [0.59 1.2 1.64]
A [0.4 0.66 0.82] [0.29 0.71 0. ] [0.49 1.52 1.96]
B [-0.48 0.66 0.88] [0. 1. 0.] [-0.57 1.65 2.04]
B [-0.87 0.63 0.93] [0. 1. 0.] [-1.49 1.7 1.97]
B [-0.96 0.61 0.92] [0. 1. 0.] [-2.41 1.72 1.77]
B [-0.97 0.61 0.89] [0. 1. 0.] [-3.32 1.73 1.53]
B [-0.97 0.61 0.82] [0. 1. 0.] [-4.22 1.74 1.23]
B [-0.97 0.61 0.68] [0. 0.99 0.01] [-5.11 1.75 0.85]
B [-0.97 0.61 0.28] [0. 0.9 0.09] [-5.98 1.76 0.3 ]
B [-0.97 0.61 -0.47] [0. 0.04 0.96] [-6.83 1.78 -0.54]
C [-0.98 0.52 -0.83] [0. 0. 1.] [-6.28 1.82 -1.26]
C [-0.98 0.52 -0.87] [0. 0. 1.] [-5.64 1.86 -1.46]
C [-0.98 0.52 -0.88] [0. 0. 1.] [-5. 1.88 -1.51]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-4.37 1.9 -1.52]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-3.74 1.92 -1.52]
C [-0.98 0.52 -0.89] [0. 0. 1.] [-3.12 1.93 -1.52]
C [-0.97 0.52 -0.89] [0. 0. 1.] [-2.51 1.94 -1.52]
C [-0.94 0.52 -0.88] [0. 0. 1.] [-1.89 1.95 -1.52]
C [-0.84 0.53 -0.88] [0. 0. 1.] [-1.28 1.96 -1.5 ]
C [-0.55 0.53 -0.87] [0. 0. 0.99] [-0.64 1.97 -1.46]
C [ 0.07 0.55 -0.84] [0.23 0.01 0.76] [ 0.08 2.01 -1.35]
C [ 0.68 0.58 -0.77] [0.95 0. 0.05] [ 0.06 2.06 -1.15]
A [0.64 0.73 0.38] [0.85 0.14 0. ] [0.91 2.34 0.63]
A [0.53 0.73 0.65] [0.57 0.43 0. ] [0.67 2.55 1.11]
A [0.42 0.72 0.76] [0.34 0.66 0. ] [0.5 2.72 1.51]
A [0.32 0.72 0.82] [0.21 0.79 0. ] [0.37 2.86 1.88]
B [-0.54 0.7 0.89] [0. 1. 0.] [-0.64 2.8 1.98]
B [-0.89 0.66 0.93] [0. 1. 0.] [-1.57 2.7 1.92]
B [-0.96 0.64 0.92] [0. 1. 0.] [-2.49 2.58 1.74]
B [-0.97 0.64 0.89] [0. 1. 0.] [-3.41 2.47 1.51]
B [-0.97 0.63 0.82] [0. 1. 0.] [-4.31 2.38 1.23]
B [-0.97 0.63 0.69] [0. 0.99 0.01] [-5.19 2.3 0.87]
B [-0.97 0.63 0.32] [0. 0.93 0.07] [-6.07 2.24 0.34]
B [-0.97 0.63 -0.43] [0. 0.06 0.94] [-6.92 2.19 -0.48]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-6.38 2.14 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-5.75 2.1 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-5.11 2.07 -1.51]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-4.48 2.05 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.85 2.03 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.23 2.02 -1.52]
C [-0.97 0.53 -0.89] [0. 0. 1.] [-2.62 2.01 -1.52]
C [-0.95 0.53 -0.89] [0. 0. 1.] [-2.01 2. -1.52]
C [-0.87 0.53 -0.88] [0. 0. 1.] [-1.39 2. -1.51]
C [-0.63 0.53 -0.88] [0. 0. 0.99] [-0.76 2. -1.48]
C [-0.06 0.54 -0.85] [0.11 0.01 0.88] [-0.06 2.02 -1.38]
C [ 0.6 0.58 -0.78] [0.92 0. 0.08] [ 0.72 2.07 -1.19]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.34 0.61]
A [0.52 0.72 0.65] [0.57 0.43 0. ] [0.66 2.56 1.1 ]
B [-0.41 0.71 0.72] [0.01 0.99 0. ] [-0.47 2.54 1.07]
B [-0.85 0.67 0.65] [0. 0.99 0.01] [-1.39 2.49 0.82]
B [-0.95 0.64 0.29] [0. 0.91 0.08] [-2.31 2.41 0.3 ]
B [-0.97 0.64 -0.46] [0. 0.05 0.95] [-3.21 2.35 -0.52]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-2.71 2.26 -1.25]
C [-0.95 0.53 -0.87] [0. 0. 1.] [-2.11 2.19 -1.45]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.5 2.14 -1.49]
C [-0.69 0.53 -0.88] [0. 0. 1.] [-0.87 2.11 -1.48]
C [-0.18 0.54 -0.86] [0.05 0.01 0.94] [-0.19 2.1 -1.4 ]
C [ 0.51 0.57 -0.8 ] [0.86 0. 0.14] [ 0.58 2.12 -1.23]
A [0.61 0.72 0.38] [0.84 0.16 0. ] [0.84 2.39 0.58]
A [0.51 0.72 0.65] [0.56 0.44 0. ] [0.65 2.59 1.09]
A [0.41 0.72 0.76] [0.34 0.66 0. ] [0.49 2.76 1.5 ]
A [0.32 0.72 0.82] [0.2 0.79 0. ] [0.37 2.89 1.88]
B [-0.54 0.7 0.89] [0. 1. 0.] [-0.65 2.82 1.99]
B [-0.89 0.66 0.93] [0. 1. 0.] [-1.58 2.72 1.92]
B [-0.96 0.64 0.92] [0. 1. 0.] [-2.5 2.6 1.74]
B [-0.97 0.64 0.89] [0. 1. 0.] [-3.41 2.49 1.51]
B [-0.97 0.63 0.83] [0. 1. 0.] [-4.31 2.39 1.24]
B [-0.97 0.63 0.69] [0. 0.99 0.01] [-5.2 2.31 0.87]
B [-0.97 0.63 0.32] [0. 0.93 0.07] [-6.07 2.25 0.34]
B [-0.97 0.63 -0.43] [0. 0.06 0.94] [-6.93 2.2 -0.48]
C [-0.98 0.53 -0.83] [0. 0. 1.] [-6.39 2.15 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-5.75 2.1 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-5.12 2.07 -1.51]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-4.49 2.05 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.86 2.03 -1.52]
C [-0.98 0.53 -0.89] [0. 0. 1.] [-3.24 2.02 -1.52]
C [-0.97 0.53 -0.89] [0. 0. 1.] [-2.62 2.01 -1.52]
C [-0.95 0.53 -0.89] [0. 0. 1.] [-2.01 2. -1.52]
C [-0.87 0.53 -0.88] [0. 0. 1.] [-1.4 2. -1.51]
C [-0.63 0.53 -0.88] [0. 0. 0.99] [-0.76 2. -1.48]
C [-0.07 0.54 -0.85] [0.1 0.01 0.89] [-0.07 2.02 -1.38]
C [ 0.6 0.57 -0.78] [0.92 0. 0.08] [ 0.72 2.07 -1.19]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.34 0.61]
A [0.52 0.72 0.65] [0.57 0.43 0. ] [0.66 2.56 1.1 ]
A [0.42 0.72 0.76] [0.34 0.66 0. ] [0.5 2.72 1.51]
B [-0.48 0.71 0.84] [0. 1. 0.] [-0.56 2.68 1.58]
B [-0.87 0.66 0.87] [0. 1. 0.] [-1.49 2.6 1.48]
B [-0.96 0.64 0.82] [0. 1. 0.] [-2.41 2.5 1.23]
B [-0.97 0.64 0.69] [0. 0.99 0.01] [-3.32 2.41 0.88]
B [-0.97 0.63 0.32] [0. 0.93 0.07] [-4.22 2.33 0.35]
B [-0.97 0.64 -0.42] [0. 0.06 0.94] [-5.1 2.27 -0.46]
C [-0.98 0.53 -0.82] [0. 0. 1.] [-4.59 2.2 -1.23]
C [-0.98 0.53 -0.87] [0. 0. 1.] [-3.97 2.15 -1.45]
C [-0.98 0.53 -0.88] [0. 0. 1.] [-3.35 2.1 -1.5 ]
C [-0.97 0.53 -0.88] [0. 0. 1.] [-2.74 2.07 -1.52]
C [-0.95 0.53 -0.89] [0. 0. 1.] [-2.12 2.05 -1.52]
C [-0.89 0.53 -0.88] [0. 0. 1.] [-1.51 2.03 -1.51]
C [-0.7 0.53 -0.88] [0. 0. 1.] [-0.88 2.03 -1.48]
C [-0.2 0.54 -0.86] [0.05 0.01 0.95] [-0.2 2.04 -1.41]
C [ 0.5 0.57 -0.8 ] [0.85 0. 0.15] [ 0.57 2.07 -1.23]
epoch: 7
loss: 0.0151
-----
```

```

-----
symbol= ABBCCCAABBBBCCCCCABBCCABBCCABBCCCA
hidden activations, output probabilities and context:
A [0.37 0.29 0.54] [0.55 0.44 0. ] [0.45 0.43 0.75]
B [-0.44 0.44 0.28] [0.04 0.9 0.06] [-0.53 0.72 0.32]
B [-0.85 0.49 -0.41] [0. 0.06 0.93] [-1.42 0.92 -0.46]
C [-0.67 0.45 -0.82] [0. 0. 0.99] [-0.83 1.16 -1.21]
C [-0.14 0.49 -0.84] [0.07 0.01 0.92] [-0.14 1.38 -1.35]
C [ 0.55 0.54 -0.79] [0.88 0. 0.11] [ 0.64 1.57 -1.21]
A [0.62 0.7 0.37] [0.85 0.15 0. ] [0.86 1.91 0.58]
A [0.52 0.71 0.64] [0.57 0.43 0. ] [0.67 2.17 1.09]
B [-0.41 0.7 0.71] [0.01 0.99 0. ] [-0.46 2.21 1.05]
B [-0.85 0.66 0.63] [0. 0.99 0.01] [-1.38 2.21 0.79]
B [-0.95 0.64 0.23] [0. 0.88 0.12] [-2.3 2.17 0.24]
B [-0.97 0.63 -0.52] [0. 0.03 0.97] [-3.21 2.14 -0.6 ]
C [-0.97 0.53 -0.84] [0. 0. 1.] [-2.69 2.1 -1.28]
C [-0.95 0.53 -0.88] [0. 0. 1.] [-2.08 2.07 -1.46]
C [-0.88 0.53 -0.88] [0. 0. 1.] [-1.47 2.05 -1.5 ]
C [-0.67 0.53 -0.88] [0. 0. 0.99] [-0.84 2.04 -1.48]
C [-0.15 0.54 -0.86] [0.06 0.01 0.93] [-0.16 2.05 -1.4 ]
C [ 0.53 0.57 -0.79] [0.88 0. 0.12] [ 0.62 2.08 -1.22]
A [0.62 0.72 0.38] [0.84 0.16 0. ] [0.85 2.35 0.59]
B [-0.33 0.71 0.22] [0.07 0.88 0.05] [-0.36 2.38 0.25]
B [-0.82 0.67 -0.42] [0. 0.08 0.92] [-1.27 2.37 -0.48]
C [-0.62 0.54 -0.82] [0. 0.01 0.99] [-0.75 2.29 -1.21]
C [-0.06 0.55 -0.84] [0.12 0.01 0.88] [-0.06 2.24 -1.33]
C [ 0.6 0.58 -0.78] [0.92 0. 0.08] [ 0.73 2.23 -1.18]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.49 0.61]
B [-0.31 0.72 0.24] [0.07 0.89 0.04] [-0.35 2.5 0.28]
B [-0.82 0.67 -0.39] [0. 0.1 0.9] [-1.26 2.47 -0.44]
C [-0.62 0.54 -0.81] [0. 0.01 0.99] [-0.74 2.37 -1.19]
C [-0.05 0.55 -0.84] [0.12 0.01 0.87] [-0.05 2.3 -1.33]
C [ 0.61 0.58 -0.78] [0.92 0. 0.07] [ 0.73 2.28 -1.18]
A [0.63 0.73 0.38] [0.85 0.15 0. ] [0.88 2.53 0.61]
B [-0.31 0.72 0.24] [0.07 0.89 0.04] [-0.35 2.53 0.28]
B [-0.82 0.67 -0.39] [0. 0.1 0.9] [-1.26 2.5 -0.43]
C [-0.62 0.54 -0.81] [0. 0.01 0.99] [-0.74 2.39 -1.19]
C [-0.05 0.55 -0.84] [0.12 0.01 0.87] [-0.05 2.32 -1.33]
C [ 0.61 0.58 -0.78] [0.92 0. 0.07] [ 0.73 2.29 -1.18]
epoch: 8
loss: 0.0056
-----

```


References

Rodriguez, P., Wiles, J., & Elman, J. L. (1999). *A recurrent neural network that learns to count*. Connection Science, 11(1), 5–40. <https://doi.org/10.1080/095400999116340>

Lecture Slides (W1 All, W2 All, W3 All, W4 All, W5 RNN)